# HYPERRIGIDITY IN LOW DIMENSIONS AND A DIMENSION-33 COUNTEREXAMPLE

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**Working draft — Version 2**
Not published or released as a preprint.

Abstract. We study the least affine dimension of a commutative counterexample to Arveson's hyperrigidity conjecture. Using an inverse-square energy estimate for supporting affine functions, we prove hyperrigidity for every compact convex subset of $\mathbb{R}^3$ with closed extreme boundary. The key step replaces part of a dilation while preserving supporting normals, after which Borel interpolation on exposed segments avoids disintegration. We also modify the Walsh dilation of Bilich–Scherer to construct a counterexample in affine dimension 33. Thus the least affine dimension lies between 4 and 33. Further results exclude several smaller construction mechanisms, including rank-one total quadratic defects, Hardy compressions of multiplicity at most three, and small polynomial perturbations of quadratic graphs. A finite-matrix criterion and connections with matrix convexity and quantum channels identify remaining directions toward dimension four.

## 1. Introduction

Let $K \subset \mathbb{R}^d$ be a compact convex set with closed extreme boundary $X = \operatorname{ex} K$. We write $A(K)$ for the complex vector space of continuous affine functions on $K$ and regard

$$A(K)|_X \subset C(X)$$

as a function system. A unital representation $\pi : C(X) \to B(H)$ has the *unique extension property* (UEP) if the only unital completely positive (UCP) map $\Phi : C(X) \to B(H)$ agreeing with $\pi$ on $A(K)|_X$ is $\pi$ itself. By Arveson's characterization, the function system is hyperrigid exactly when every unital representation has this property [Arv]. We use this formulation throughout.

Each point evaluation at an extreme point has the unique extension property: a probability measure with that barycenter is a point mass. The question is whether the same uniqueness survives for diffuse representations. Bilich and Scherer showed that it need not [BS]. Their construction raises a geometric question: how small can the affine dimension of a commutative counterexample be?

The present paper proves universal hyperrigidity in dimension three, gives an explicit counterexample in dimension 33, and structural restrictions on attempts to bridge the gap. Its first main result is the following.

*Date*: Version 2, 23 October 2026.

*Key words and phrases.* Hyperrigidity, unique extension property, compact convex sets, positive operator-valued measures, Riesz energy.

Prepared with substantial AI assistance and automated adversarial review. Independent human verification and a complete novelty assessment remain pending; no formal verification is claimed.

**Theorem A.** *Let $K \subset \mathbb{R}^3$ be a compact convex set with closed extreme boundary. Then $A(K)|_{\operatorname{ex} K}$ is hyperrigid in $C(\operatorname{ex} K)$.*

No smoothness or exposedness hypothesis is imposed. The theorem extends the positive results in the first version of this paper. The new proof retains every exceptional source point: countably many simple pieces are handled separately, and the residual points are endpoints of exposed segments. A dilation replacement preserves their supporting normals, allowing bounded Borel interpolation between the segment endpoints.

The common mechanism is an energy estimate. If $n_x$ and $n_y$ are unit outward normals at $x, y \in \partial K$, then

$$0 \le \langle n_x, x - y \rangle \le \frac{1}{4}\big|(x + n_x) - (y + n_y)\big|^2. \tag{1.1}$$

The normal lift $x \mapsto x + n_x$ takes values on $\partial(K + \overline{B})$, where $\overline{B}$ is the closed unit ball. In dimension three, every nonzero measure on this parallel surface has infinite inverse-square energy. On the other hand, a nonzero separated block of an agreeing positive operator-valued measure would produce a finite-energy measure. This contradiction proves the required separation. The replacement step assigns each new target copy its source normal, even when several sources share a nonsmooth target. The complete proof is given in Theorem 5.1.

Our second main result is constructive.

**Theorem B.** *There is a compact convex body $K \subset \mathbb{R}^{33}$ with closed extreme boundary $X$, a unital representation $\pi : C(X) \to B(H)$, and a unital completely positive map $\Phi : C(X) \to B(H)$ such that*

$$\Phi|_{A(K)|_X} = \pi|_{A(K)|_X}, \qquad \Phi \neq \pi.$$

*The Hilbert space $H$ is separable and infinite dimensional, and the spectral measure of $\pi$ has no point atoms. Moreover $\dim_{\mathbb{C}} A(K) = 34$.*

The construction is a change of distribution in the Walsh dilation of Bilich–Scherer, rather than a new dilation mechanism. Theorem 6.1 gives the complete data. The operator identities are proved before the numerical parameters are chosen; the remaining containment estimate is a scalar inequality for every infinite digit sequence. A continuous barycentric kernel then gives the extension, and a quadratic function provides an explicit positive witness of disagreement.

For comparison, [BS, Corollary 3.3] proves a construction in ambient dimension 129, states an optimization to dimension 57, and mentions an unverified dimension-11 proposal in Remark 3.4. Scherer's planar theorem [Sch1, Theorem 3.8] excludes dimensions at most two. Together, Theorems A and B give

$$4 \le d_{\min} \le 33, \tag{1.2}$$

where $d_{\min}$ denotes the least affine dimension under the closed-extreme-boundary hypothesis. Whether the lower bound four is attained remains open. The positive theorem for compact spectrahedra [Sch2, Theorem 3.7] is a further point of comparison; it applies to finite self-adjoint affine matrix pencils, not arbitrary projections of spectrahedra.

The construction obstructions explain why simply changing its visible geometry does not close this gap. A finite-dimensional signing lemma rules out the quadratic-distance certificate for every stationary three-point distribution and every geometric-norm vector encoding in horizontal dimension at most four, under the nonnegative-potential and exact-height assumptions of Theorem 7.2. Thus even arbitrary directions cannot make that certificate work in affine dimension four or five. A stronger geometric argument rules out *every finite strictly convex graph* for the same stationary mechanism in affine dimension four (Corollary 8.6). It uses positive

distance levels of the represented set and requires no differentiability of the graph. These are obstructions to a specified construction family, not universal lower bounds on $d_{\min}$.

The general supporting-gap criterion, Theorem 3.1, is useful beyond these applications. Appendix A gives a source in a four-dimensional convex body whose normal lift supports a finite-energy measure, but whose representations still have the unique extension property by a sharper metric. Appendices B and C treat varying and correlated distributions. Appendix D records two restrictions on direct operator constructions. The later sections add independent construction obstructions and a finite-POVM criterion, followed by a literature-guided discussion of dimension four. Exploratory numerical searches and superseded certificates are not needed for any proof in this paper.

## 2. Representations, measures, and defects

All Hilbert spaces are complex. Representations and completely positive maps are unital unless stated otherwise. A convex body has nonempty interior in its ambient Euclidean space. We write $\overline{B}$ for the closed unit ball and use $|\cdot|$ for Euclidean norms. For a compact convex set $K$, its support function is $h_K(u) = \max_{x\in K}\langle u, x\rangle$. An exposed point is the unique minimizer of an affine function on $K$. A smooth boundary point has a unique unit outward normal. A strictly convex body has no nondegenerate segment in its boundary; every boundary point is then exposed.

Whenever $X = \operatorname{ex} K$ is compact, the coordinate functions and the constant function generate $C(X)$. If $\operatorname{aff} K = \mathbb{R}^d$, their complex linear span is $A(K)|_X$ and has dimension $d+1$. The dimension of a representation space is a separate issue.

For a representation $\pi : C(X) \to B(H)$ we write $P$ for its spectral measure. A UCP map $\Phi$ has a positive operator-valued measure $E$. One may construct its bounded Borel extension from a Stinespring representation $\Phi(f) = V^*\rho(f)V$: if $Q$ is the spectral measure of $\rho$, set $E(B) = V^*Q(B)V$. All operator-valued measures below are countably additive in the strong operator topology. Saying that a representation is concentrated on a Borel set $Z$ means $P(Z) = I$; this is stronger than an assertion about the closure of that set.

**Lemma 2.1** (Coordinate defects)**.** *Suppose $\Phi = V^*\rho(\,\cdot\,)V$ agrees with $\pi$ on the affine coordinates $x_1, \dots, x_d$. Then*

$$D_j := \Phi(x_j^2) - \pi(x_j)^2 = \big[(I - VV^*)\rho(x_j)V\big]^*\big[(I - VV^*)\rho(x_j)V\big] \geq 0.$$

*All $D_j$ vanish if and only if $VH$ reduces $\rho(C(X))$, and this is equivalent to $\Phi = \pi$.*

*Proof.* Expand the square and use $V^*\rho(x_j)V = \pi(x_j)$. If it vanishes, $VH$ is invariant for the self-adjoint operator $\rho(x_j)$ and hence reduces it. The coordinates generate $C(X)$, so the compression is a representation agreeing with $\pi$ on a generating set. The converse follows from the same identity. □

**Lemma 2.2** (Atomic representations)**.** *Every Hilbert direct sum of point evaluations at points of $X$, with arbitrary multiplicities, has the unique extension property.*

*Proof.* Write $H = \bigoplus_x H_x$, with $\pi(f)|_{H_x} = f(x)I$. For a unit vector $h \in H_x$, the scalar measure representing $f \mapsto \langle \Phi(f)h, h\rangle$ has barycenter $x$ and is therefore $\delta_x$. Indeed, a probability measure with an extreme barycenter cannot be split into two parts with different conditional barycenters. In a Stinespring representation this gives

$$\|\rho(f)Vh - f(x)Vh\|^2 = \langle \Phi(|f - f(x)|^2)h, h\rangle = 0.$$

Thus $\rho(f)V = V\pi(f)$ on each summand and then on their Hilbert direct sum. This proves equality, including all off-diagonal coefficients. □

In particular, finite-dimensional representations cannot witness a counterexample. We shall also need a stronger local consequence of extremality.

**Lemma 2.3** (Local vanishing of the defect)**.** *Let $\Phi$ agree with $\pi$ on $A(K)|_X$, and let $D_j$ be the defects above. If $x$ belongs to the spectral support of $\pi$, then*

$$\lim_{r\downarrow 0}\left\| P(B(x,r))\left(\sum_j D_j\right)P(B(x,r))\right\| = 0.$$

*Consequently no coordinate defect, nor their sum, is bounded below by a positive multiple of the identity.*

*Proof.* Any sequence of probability measures on $X$ whose barycenters tend to $x$ converges weakly to $\delta_x$: every weak limit has the extreme barycenter $x$. By compactness, their second moments about $x$ tend to zero uniformly as their barycenters approach $x$. For a unit vector $h \in P(B(x,r))H$, the scalar measure of $\Phi$ at $h$ has barycenter within $r$ of $x$. Affine agreement gives

$$\left\langle \sum_j D_j h, h\right\rangle = \int_X |y-x|^2\, d\mu_h(y) - \left\langle \pi(|\,\cdot - x|^2)h, h\right\rangle \le \int_X |y-x|^2\, d\mu_h(y).$$

The right side tends to zero uniformly in these unit vectors. Positivity gives the norm estimate. Every such spectral neighborhood is nonzero, which proves the last assertion. □

## 3. A quadratic affine-barrier criterion

The positive results in this section rest on a common observation. An affine supporting function can bound the amount of a dilation that passes from one part of the spectrum to another. If these bounds vanish quadratically in a suitable metric, a nonzero off-diagonal effect would produce a scalar measure of finite inverse-square energy. A critical covering estimate rules out such a measure. The metric need not be the ambient Euclidean metric, and this flexibility will be useful below.

We retain the notation of Section 2: $P$ is the spectral measure of $\pi$, and $E$ is the POVM of an affine-agreeing UCP map $\Phi$. Their bounded Borel extensions will be used without changing notation.

**Theorem 3.1** (Quadratic affine barriers)**.** *Let $\pi : C(X) \to B(H)$ be a unital representation, and let $Z \subset X$ be Borel with $P(Z) = I$. Suppose that, for each $x \in Z$, an affine function $b_x$ on $K$ is specified so that its coefficients depend Borel measurably on $x$ and*

$$b_x(x) = 0, \qquad b_x(y) > 0 \quad (y \in K \setminus \{x\}).$$

*Suppose also that there are a compact metric space $(T, d_T)$, a Borel map $\theta : Z \to T$, and a finite constant $M$ such that*

$$b_x(y) \le M\, d_T(\theta(x),\theta(y))^2 \qquad (x, y \in Z).$$

*Assume that every nonzero finite positive Borel measure $\mu$ on $Z$ satisfies*

$$\iint_{Z\times Z} d_T(\theta(x),\theta(y))^{-2}\, d\mu(x)d\mu(y) = +\infty,$$

*where the integrand is $+\infty$ when the distance is zero. Then $\pi$ has the unique extension property relative to $A(K)|_X$.*

We first establish the analytic facts needed for the proof. For a finite positive measure $\nu$ on a metric space, write

$$I_2(\nu) = \iint d(u,v)^{-2}\, d\nu(u)d\nu(v).$$

A Borel set has *zero inverse-square capacity* here if every nonzero finite positive measure concentrated on it has infinite $I_2$-energy. The formulation in Theorem 3.1 is the corresponding statement for the pushforwards under $\theta$; it does not require a separate measurability assertion about the image of $\theta$.

### 3.1. **Energy and localization.**

**Lemma 3.2** (Rank-one domination)**.** *Let $P$ be a spectral measure on a compact metric space, let $w$ be a bounded nonnegative Borel function, and suppose*

$$|\eta\rangle\langle\eta| \le M \int w\, dP.$$

*Then, for $\mu_\eta(D) = \langle P(D)\eta, \eta\rangle$,*

$$\int w^{-1}\, d\mu_\eta \le M,$$

*with $w^{-1} = +\infty$ on the zero set of $w$.*

*Proof.* Put $W = \int w\, dP$. For $\epsilon > 0$, set $h = (W + \epsilon I)^{-1}\eta$ and $t = \langle \eta, h\rangle \ge 0$. The assumed domination gives

$$t^2 \le M\langle Wh, h\rangle \le M\langle (W + \epsilon I)h, h\rangle = Mt.$$

Thus $t \le M$. The spectral theorem and monotone convergence as $\epsilon \downarrow 0$ prove the assertion. □

**Lemma 3.3** (Critical covering bound)**.** *Let $T$ be a compact metric space. Suppose that, for every sufficiently small $\epsilon > 0$, it can be covered by at most $M\epsilon^{-2}$ sets of diameter at most $\epsilon$. Then $I_2(\nu) = +\infty$ for every nonzero finite positive Borel measure $\nu$ on $T$. In particular, this conclusion holds for the boundary of every compact convex body in $\mathbb{R}^3$.*

*Proof.* The covering sets may be replaced by their closures and then refined to a Borel partition $T = \bigsqcup_{j=1}^{N_\epsilon} T_j$, with $N_\epsilon \le M\epsilon^{-2}$ and $\operatorname{diam} T_j \le \epsilon$. If $m = \nu(T) > 0$, Cauchy–Schwarz gives

$$(\nu \times \nu)\{d(u, v) \le \epsilon\} \ge \sum_j \nu(T_j)^2 \ge \frac{m^2}{N_\epsilon} \ge \frac{m^2}{M}\epsilon^2.$$

The layer-cake formula and Tonelli's theorem therefore imply

$$I_2(\nu) \ge 2\int_0^{\epsilon_0} t^{-3}(\nu \times \nu)\{d(u, v) \le t\}\, dt \ge \frac{2m^2}{M}\int_0^{\epsilon_0} \frac{dt}{t} = +\infty.$$

For the final assertion, translate the convex body so that $r\overline{B}_3 \subset K \subset R\overline{B}_3$ for some $r, R > 0$. Its Minkowski functional $j_K$ is $1/r$-Lipschitz and is at least $1/R$ on $S^2$. Hence the radial map $u \mapsto u/j_K(u)$ is Lipschitz from $S^2$ onto $\partial K$. The sphere admits the required $O(\epsilon^{-2})$ covers, as follows from finitely many Lipschitz coordinate charts. The radial map transfers this covering bound to $\partial K$. □

The next lemma isolates the operator argument. Unlike Theorem 3.1, it does not require each affine function to expose a single point.

**Lemma 3.4** (Local annihilation)**.** *Suppose $\Phi$ and $\pi$ agree on $A(K)|_X$. Let $F, G \subset X$ be Borel. For each $x \in F$, let $b_x$ be an affine function nonnegative on $K$. Suppose that*

$$b_x(y) \ge \delta > 0 \quad (x \in F,\ y \in G), \qquad b_x(y) \le M\, d_T(\theta(x), \theta(y))^2 \quad (x, y \in F),$$

*where $\theta : F \to T$ is Borel and $T$ is compact metric. If every nonzero finite positive measure on $F$ has infinite energy in the displayed metric, then $E(G)P(F) = 0$.*

*Proof.* Set $A = P(F)E(G)P(F)$. For each fixed $x \in F$, positivity and affine agreement imply

$$0 \le A \le \delta^{-1}P(F)\Phi(b_x)P(F) = \delta^{-1}P(F)\pi(b_x)P(F) \le \frac{M}{\delta}\int_F d_T(\theta(x),\theta(y))^2\,dP(y).$$

If $A \neq 0$, choose a unit vector $h$ such that $\eta = A^{1/2}h \neq 0$. Then $|\eta\rangle\langle\eta| \le A$ and $\eta \in P(F)H$. Lemma 3.2, applied on $P(F)H$, gives

$$\int_F d_T(\theta(x),\theta(y))^{-2}\,d\mu_\eta(y) \le M/\delta \qquad (x \in F).$$

The same vector $\eta$ works for every $x \in F$. Tonelli's theorem therefore gives

$$\iint_{F\times F} d_T(\theta(x),\theta(y))^{-2}\,d\mu_\eta(y)d\mu_\eta(x) \le (M/\delta)\|\eta\|^2 < \infty,$$

contrary to the energy hypothesis. Thus $A = 0$. Since $A = (E(G)^{1/2}P(F))^*(E(G)^{1/2}P(F))$, it follows that $E(G)P(F) = 0$. □

We will also use the following elementary passage from compact sets to the full spectral measure.

**Lemma 3.5.** *Let $P$ be a normalized spectral measure and $E$ a normalized POVM on a compact metric space. If $E(B)P(A) = 0$ whenever $A, B$ are disjoint compact sets, then $E = P$.*

*Proof.* For an open set $O$, choose compact sets $A_j \uparrow O$. Applying the hypothesis in both orders to $A_j$ and the compact set $X \setminus O$, and taking strong limits, gives

$$E(X \setminus O)P(O) = 0, \qquad E(O)P(X \setminus O) = 0.$$

The two effects sum to $I$, so $E(O)P(O) = P(O)$ and $E(O)P(X \setminus O) = 0$. Hence $E(O) = P(O)$. The scalar matrix-coefficient measures are regular and agree on open sets; they therefore agree on all Borel sets. □

*Proof of Theorem 3.1.* Let $\Phi$ be an agreeing UCP map. Fix disjoint nonempty compact sets $A, B \subset X$, and put

$$F_m = \left\{x \in A \cap Z : \min_{y\in B} b_x(y) \ge 1/m\right\}.$$

The minimum over $B$ is a continuous function of the affine coefficients, so $F_m$ is Borel. Strict positivity away from $x$ and compactness of $B$ show that $F_m \uparrow A \cap Z$. Lemma 3.4 gives $E(B)P(F_m) = 0$. Since $P(Z) = I$, strong convergence yields $E(B)P(A) = 0$. Lemma 3.5 now gives $E = P$, and hence $\Phi = \pi$. □

3.2. **A graph consequence.** The criterion applies to represented subsets, independently of the size of the rest of the extreme boundary. The following consequence is useful when a construction begins with a convex graph.

**Corollary 3.6.** *Let $B \subset \mathbb{R}^D$ be compact and convex, and let $g$ be strictly convex and $C^{1,1}$ on an open convex neighborhood of $B$. Put*

$$K = \{(a,z) : a \in B,\ g(a) \le z \le h\}, \qquad h > \max_B g,$$

*and assume $\operatorname{ex} K$ is closed. Let $S \subset B$ be compact and admit covers by $O(\epsilon^{-2})$ sets of diameter at most $\epsilon$. Every representation concentrated on*

$$Z = \{(s, g(s)) : s \in S\}$$

*has the unique extension property.*

*Proof.* For $x = (s, g(s))$, take the affine barrier

$$b_x(a, z) = z - g(s) - \langle \nabla g(s), a - s \rangle.$$

Strict convexity makes this nonnegative on $K$, with equality only at $x$. A Lipschitz bound $L$ for $\nabla g$ on $B$ gives

$$b_{(s,g(s))}(t, g(t)) \le \tfrac{L}{2}|s - t|^2 \qquad (s, t \in S).$$

Use $\theta(s, g(s)) = s$. Lemma 3.3 and Theorem 3.1 give the conclusion. □

## 4. Normal lifts and local separation

Supporting normals provide a canonical metric for the barrier criterion. The useful feature is that they can be added to the boundary points: this places the resulting metric space on the boundary of a parallel convex body. In dimension three, that surface has precisely the critical covering bound of Lemma 3.3.

For $x \in K$, let

$$N_K(x) = \{u \in \mathbb{R}^d : \langle u, y - x \rangle \le 0 \text{ for every } y \in K\}$$

be the outward normal cone. For a unit vector $n_x \in N_K(x)$, put $z_x = x + n_x$. The point $z_x$ lies on $\partial(K + \overline{B}_d)$, and its metric projection onto $K$ is $x$. Indeed, for $y \in K$,

$$|x + n_x - y|^2 = 1 + |x - y|^2 + 2\langle n_x, x - y \rangle \ge 1,$$

with equality only when $y = x$. For any two such choices,

$$0 \le \langle n_x, x - y \rangle \le \langle n_x - n_y, x - y \rangle \le \tfrac{1}{4}|z_x - z_y|^2. \tag{4.1}$$

The middle inequality uses the supporting property at $y$; the last follows from $4\langle a, b \rangle \le |a + b|^2$.

**Corollary 4.1** (Normal-lift criterion)**.** *Let $\pi : C(X) \to B(H)$ have spectral measure concentrated on a Borel set $Z \subset X$. Suppose a Borel unit normal $n(x)$ exposes $x$ for every $x \in Z$. If every nonzero finite positive Borel measure concentrated on*

$$L = \{x + n(x) : x \in Z\}$$

*has infinite inverse-square energy, then $\pi$ has the unique extension property.*

*Proof.* The map $x \mapsto x + n(x)$ is Borel and injective. Its image is Borel as well: if $p_K$ is the continuous metric projection onto $K$, then

$$L = \{z \in \partial(K + \overline{B}_d) : p_K(z) \in Z,\ z - p_K(z) = n(p_K(z))\}.$$

Apply Theorem 3.1 with $b_x(y) = \langle n(x), x - y \rangle$ and $\theta(x) = x + n(x)$. The quadratic bound is (4.1), and the energy assumption passes to pushforward measures. □

We need a local version in which the normal need not expose the source point. This allows flat supporting faces in the general three-dimensional argument of the next section.

**Lemma 4.2** (Separated normal selections)**.** *Let $K \subset \mathbb{R}^3$ be a compact convex body, and suppose $\Phi$ and $\pi$ agree on $A(K)|_X$. Let $F, G \subset X$ be Borel. If there is a Borel unit normal selection $x \mapsto n_x$ on $F$ such that*

$$\langle n_x, x - y \rangle \ge \delta > 0 \qquad (x \in F,\ y \in G),$$

*then $Q(G)VP(F) = 0$ and $E(G)P(F) = 0$.*

*Proof.* Use the affine functions $b_x(y) = \langle n_x, x - y\rangle$ in Lemma 3.4. By (4.1), their values on $F$ are bounded by one quarter of the squared distance between the normal lifts. Every nonzero pushforward measure on these lifts has infinite energy, since the lifts lie on $\partial(K + \overline{B}_3)$ and Lemma 3.3 applies to that surface. The conclusion $E(G)P(F) = 0$ also gives $Q(G)VP(F) = 0$ because the square of its norm on a vector is the corresponding quadratic form of $P(F)E(G)P(F)$. □

For exposed points, a separating selection can be made on successively larger compact sets. We record the elementary selection argument to make the measurability explicit.

**Lemma 4.3** (Compact-graph selection)**.** *Let $F, T$ be compact metric spaces, and let $\Gamma \subset F \times T$ be compact with every fiber $\Gamma_x$ nonempty. There is a Borel map $s : F \to T$ with $(x, s(x)) \in \Gamma$ for every $x \in F$.*

*Proof.* Fix a countable dense sequence in $T$. At step $j$, choose the first closed ball of radius $2^{-j}$ centered in that sequence which meets the current nonempty compact fiber, and intersect the fiber with that ball. These choices are Borel. Indeed, for each fixed finite list of earlier choices, the nonemptiness test is the projection of the original compact graph intersected with finitely many fixed closed balls; this projection is compact. The possible earlier lists form a countable Borel partition, so induction proves the claim. The remaining fibers are nested nonempty compact sets with diameters tending to zero. Their unique intersection point is the limit of the selected centers, and hence depends Borel measurably on $x$. □

## 5. Hyperrigidity in affine dimension three

**Theorem 5.1.** *Let $K \subset \mathbb{R}^3$ be a compact convex body and assume that $X = \operatorname{ex} K$ is closed. Every unital representation of $C(X)$ has the unique extension property relative to $A(K)|_X$. Consequently, $A(K)|_X$ is hyperrigid in $C(X)$.*

Write $F_u = \{y \in K : \langle u, y\rangle = h_K(u)\}$ for $u \in S^2$. We retain the normal-cone notation $N_K(x)$ from the preceding section. A boundary point is smooth when $N_K(x) \cap S^2$ is a singleton. Throughout, $\pi : C(X) \to B(H)$ is a representation and $\Phi(f) = V^*\rho(f)V$ is a UCP map agreeing with $\pi$ on affine functions. Here $V : H \to L$ is an isometry and $\rho : C(X) \to B(L)$ is a unital representation. Their spectral measures are $P, Q$, and $E(B) = V^*Q(B)V$. Bounded Borel functional calculus is understood.

**Lemma 5.2** (Separable reduction)**.** *To prove uniqueness for an affine-agreeing pair $(\pi, \Phi)$, or an identity $Q(B)VP(A) = 0$, it suffices to treat separable source and Stinespring spaces. Borel spectral concentration assumptions persist.*

*Proof.* Fix a vector witnessing failure of the asserted identity. The C*-algebra generated by $\pi(C(X))$ and $\Phi(C(X))$ is separable because $C(X)$ is separable. The closed orbit of this vector under that algebra is a separable reducing subspace $H_0$ for both maps. The space $L_0 = \overline{\operatorname{span}\, \rho(C(X))VH_0}$ is separable, reduces $\rho$, and contains $VH_0$; it supplies the restricted Stinespring representation. Reducing subspaces for a representation also reduce its bounded Borel functional calculus. Hence $P(A)H_0 \subset H_0$ and $Q(B)L_0 \subset L_0$, so either type of witness is retained. The same observation preserves all Borel spectral concentration assumptions. □

### 5.1. Contact localization.

**Lemma 5.3** (Borel contact localization)**.** *Let $S \subset X$ be the smooth locus and $n : S \to S^2$ its unique normal map. If Borel sets $A \subset S$ and $B \subset X$ satisfy*

$$\langle n(x), x - y\rangle > 0 \qquad (x \in A,\ y \in B),$$

*then $Q(B)VP(A) = 0$.*

*Proof.* By Lemma 5.2, assume the representation spaces are separable. A positive summable combination of scalar spectral measures from countable dense sets of unit vectors gives a finite scalar measure $\nu$ with respect to which every scalar spectral measure of $P, Q$ is absolutely continuous. Increasing compact subsets of $A$ and $B$ exhaust each set modulo a $\nu$-null set. The normal map is continuous relative to $S$: the unit-normal correspondence has closed graph and compact values, so a limit of unique normals at a smooth limiting point must be its unique normal. On each product of these compact sets the displayed gap has a positive minimum. Lemma 4.2 applies, and successive strong limits give the claim. □

5.2. **The source decomposition.**

**Lemma 5.4.** *For a compact convex body in $\mathbb{R}^3$, the nonexposed nonsmooth extreme points form a countable set. Its two-dimensional exposed faces also form a countable collection.*

*Proof.* Let $x$ be nonexposed and nonsmooth. Choose linearly independent $u_1, \dots, u_k \in N_K(x)$ spanning $\operatorname{span} N_K(x)$ and put $u = (\sum_i u_i)/|\sum_i u_i|$. The cone is pointed, so the denominator is nonzero. Nonnegativity of the supporting losses implies

$$F_u = \bigcap_{v \in N_K(x)} \{y \in K : \langle v, x - y\rangle = 0\}.$$

This face has dimension at most $3 - k$. Nonsmoothness gives $k \geq 2$, whereas nonexposedness gives $\dim F_u \geq 1$. Hence $k = 2$ and $F_u$ is an exposed segment with $x$ an endpoint. For an interior point $p$ of that segment, $N_K(p) = N_K(x)$: all normals at $x$ vanish on the segment, and a normal at an interior point supports both endpoints.

There are at most countably many exposed segments $F$ having a two-dimensional normal cone in their relative interiors. To see this, choose a nondegenerate closed subsegment $F_0 \subset \operatorname{relint} F$ and a nondegenerate closed arc $C_0 \subset N_K(F) \cap S^2$, where $N_K(F)$ denotes the common normal cone on $\operatorname{relint} F$. The patch $F_0 + C_0$ lies on $\partial(K + \overline{B}_3)$ and has positive area: the segment direction and the unit-speed circular-arc direction are orthogonal, so its area is the product of their lengths. Different segments give disjoint patches, since metric projection onto $K$ sends each patch into the relative interior of its own segment. The parallel surface has finite area, so only countably many disjoint positive-area patches are possible. Each segment has two endpoints.

Finally, the relative interiors of distinct two-dimensional exposed faces are disjoint positive-area Borel subsets of $\partial K$, which also has finite area. Thus these faces are countable. □

**Lemma 5.5.** *The smooth locus $S$ and the exposed-point set $\mathcal{E}$ in $X$ are Borel. Let $\{F_j\}$ list the two-dimensional exposed faces, and put*

$$Z = X \setminus \left( \mathcal{E} \cup C \cup \bigcup_j F_j \right), \qquad C = \{\textit{nonexposed nonsmooth extreme points}\}.$$

*Then $Z$ is Borel, every $x \in Z$ is smooth, and*

$$F_{n(x)} = [x, \tau(x)]$$

*for a unique other endpoint $\tau(x) \in X \setminus \{x\}$. The map $\tau : Z \to X$ is Borel.*

*Proof.* The compact-valued unit-normal correspondence has closed graph, so its diameter is upper semicontinuous. Its singleton locus $S$ is Borel. The face correspondence $u \mapsto F_u$ has compact graph, and $u \mapsto \operatorname{diam} F_u$ is upper semicontinuous. Thus the smooth exposed points are exactly the Borel set $\{x \in S : \operatorname{diam} F_{n(x)} = 0\}$. The remaining exposed points are $(X \setminus S) \setminus C$, a Borel set because $C$ is countable by Lemma 5.4.

For $x \in Z$, its supporting face cannot have dimension zero or two. It is therefore a segment, with $x$ one endpoint because $x$ is extreme. The other endpoint is extreme in that face and hence

in $K$. For completeness, measurability of this endpoint needs no selection theorem. If $D \subset K$ is compact, define

$$m_D(x) = \max\{|y - x| : y \in F_{n(x)} \cap D\},$$

with value $-\infty$ for an empty intersection. Closedness of the face graph and compactness of $D$ show that $m_D$ is upper semicontinuous relative to $Z$. The point $\tau(x)$ is the unique maximizer defining $m_K(x)$. For an open set $O \subset K$,

$$\tau(x) \in O \quad \Longleftrightarrow \quad m_{K\setminus O}(x) < m_K(x).$$

Both functions are Borel; hence $\tau$ is Borel. □

**Lemma 5.6** (Borel endpoints of exposed segments)**.** *Let $K \subset \mathbb{R}^3$ be a compact convex body. The set*

$$U = \{u \in S^2 : F_u \text{ is a nondegenerate segment}\}$$

*is Borel. There are Borel maps $a, b : U \to \operatorname{ex} K$ such that $F_u = [a(u), b(u)]$ and $a(u) \neq b(u)$ for every $u \in U$.*

*Proof.* The face correspondence has compact graph. Consequently its diameter and the maximal area of a triangle with vertices in $F_u$ are upper semicontinuous functions of $u$. The set $U$ is characterized by positive diameter and zero maximal triangle area, so it is Borel.

For $1 \leq j \leq 3$, put

$$\alpha_j(u) = \min_{x \in F_u} x_j, \qquad \beta_j(u) = \max_{x \in F_u} x_j.$$

The functions $\alpha_j$ are lower semicontinuous and $\beta_j$ are upper semicontinuous. Partition $U$ into the Borel sets $U_j$ on which $j$ is the first coordinate satisfying $\beta_j(u) > \alpha_j(u)$. For $u \in U_j$, let $a(u)$ and $b(u)$ be the unique points of $F_u$ minimizing and maximizing that coordinate. They are its two endpoints, so they belong to $\operatorname{ex} K$.

Here is a direct measurability check, requiring no selection theorem. For compact $D \subset K$, the extended-valued function

$$\alpha_{D,j}(u) = \min\{x_j : x \in F_u \cap D\}, \qquad \min \varnothing = +\infty,$$

is lower semicontinuous. Indeed, along any convergent sequence of normals whose corresponding minima have a finite lower limit, compactness provides a convergent subsequence of minimizers, and the face graph is closed. For any relatively open $O \subset K$ and $u \in U_j$,

$$a(u) \in O \quad \Longleftrightarrow \quad \alpha_{K\setminus O,j}(u) > \alpha_j(u).$$

The equivalence uses uniqueness of the minimizing endpoint and compactness of $F_u \cap (K \setminus O)$. The right-hand side is Borel. Thus $a$ is Borel on each $U_j$. Applying the same argument to maxima, with $\max \varnothing = -\infty$, proves the assertion for $b$. □

**Lemma 5.7.** *Every representation concentrated on $\mathcal{E}$, on a countable subset of $X$, or on $X \cap F$ for a single proper exposed face $F$ has the unique extension property. A countable Borel partition into sets with this property also has this property.*

*Proof.* By Lemma 5.2, it is enough to work in separable spaces, where compact approximation uses a common finite scalar control measure as above. For exposed sources, first fix disjoint compact sets $A \subset \mathcal{E}$ and $B \subset X$. For each positive integer $m$, consider the compact graph

$$\Gamma_m = \{(x, u) \in A \times S^2 : u \in N_K(x),\ \min_{y \in B} \langle u, x - y\rangle \geq 1/m\}.$$

Its compact projection $A_m$ increases to $A$. A Borel selection of its nonempty compact fibers is supplied by Lemma 4.3. Lemma 4.2 and a strong limit give $E(B)P(A) = 0$. If $P(\mathcal{E}) = I$, compact approximation of $A_0 \cap \mathcal{E}$ for arbitrary disjoint compact $A_0, B_0 \subset X$ gives $E(B_0)P(A_0) = 0$. For

an open set $O$, exhaust $O$ by compact sets and use the two orders of the preceding annihilation with the compact set $X \setminus O$. Strong limits give

$$E(O)P(X \setminus O) = 0, \qquad E(X \setminus O)P(O) = 0.$$

Since the two effects sum to $I$, these identities imply $E(O) = P(O)$. Regularity of the scalar matrix-coefficient measures gives $E = P$.

For an atom $x$, every scalar representing measure with barycenter $x$ is $\delta_x$, since $x$ is extreme. This gives the unique extension property for evaluation at $x$ with any multiplicity. Countable atomic representations are direct sums of these.

If $F = \{b = 0\}$ for an affine $b \geq 0$ on $K$ and $P(F) = I$, then $V^*\rho(b)V = \pi(b) = 0$. Thus $Q(F)V = V$ and the dilation restricts to $C(X \cap F)$. The identity $X \cap F = \operatorname{ex} F$ and compactness of this set allow Scherer's planar theorem [Sch1, Theorem 3.8] to be applied; every affine function on $F$ extends to the ambient space.

Finally, a representation is a direct sum over a countable Borel partition. If the compression to a summand $H_j$ is equal to its representation on all of $C(X)$, the Stinespring isometry $V_j = V|_{H_j}$ satisfies $\rho(f)V_j = V_j\pi_j(f)$. This follows, for example, by expanding the square of the difference using the equalities for $f^*f, f, f^*$. Combining these intertwinings over the orthogonal sum proves the assertion. □

### 5.3. **The dilation replacement and endpoint interpolation.**

**Proposition 5.8.** *Every representation concentrated on $Z$ from Lemma 5.5 has the unique extension property.*

*Proof.* By Lemma 5.2, assume the source and target spaces are separable.

Put $Y = X \setminus S$ and split the target space using its actual spectral measure:

$$V_s = Q(S)V, \qquad V_y = Q(Y)V, \qquad A = V_y^*V_y.$$

Write $\rho_s$ and $\rho_y$ for the restrictions of $\rho$ to these reducing target spaces. These are contractions $V_s, V_y$, not generally isometries.

*Normal intertwining for the smooth target part.* Let $P_n(D) = P(Z \cap n^{-1}(D))$ and $Q_{s,n}(D) = Q(S \cap n^{-1}(D))$. If $x \in Z$ and $y \in S$ have distinct normals, their contact gap is strictly positive: equality would make $n(x)$ a unit normal at the smooth point $y$. Apply Lemma 5.3 to source and target sets with normals in $D$ and $D^c$, respectively, and then reverse them. For every Borel $D \subset S^2$ this gives

$$Q_{s,n}(D)V_s = V_sP_n(D).$$

Consequently $V_s^*V_s$ commutes with $P_n(D)$, and so does $A = I - V_s^*V_s$.

*A graph intertwiner for the nonsmooth target part.* For $x \in Z$ and $y \in Y$, the equality $\langle n(x), x - y\rangle = 0$ holds precisely when $y = \tau(x)$. Indeed, $X \cap F_{n(x)} = \{x, \tau(x)\}$ and $x \notin Y$. For a Borel set $B \subset X$, apply Lemma 5.3 to

$$(Z \cap \tau^{-1}(B^c),\ Y \cap B) \quad \text{and} \quad (Z \cap \tau^{-1}(B),\ Y \setminus B).$$

Both products have strictly positive contact gap. The resulting two annihilations give the exact graph relation

$$Q(B)V_y = V_yP(Z \cap \tau^{-1}(B)).$$

Thus, defining a unital representation on the original source space by

$$\pi_\tau(f) = \int_Z f(\tau(x))\, dP(x),$$

bounded Borel functional calculus gives

$$\rho_y(f)V_y = V_y\pi_\tau(f), \qquad V_y^*\rho_y(f)V_y = A\pi_\tau(f) = \pi_\tau(f)A.$$

Here bounded Borel functions on $Z$, such as $f \circ \tau$, may be extended arbitrarily to $X \setminus Z$; the resulting operators are independent of that extension because $P(Z) = I$. In particular $A$ commutes with $\pi_\tau(C(X))$ as well as with the normal algebra $P_n$. The formula remains valid when $\tau(x)$ is smooth: the graph relation simply makes $V_y$ vanish on that source part.

*Replacement of the target.* On $Q(S)L \oplus H$ define

$$\widetilde{\rho}(f) = \rho_s(f) \oplus \pi_\tau(f), \qquad Wh = V_s h \oplus A^{1/2} h.$$

Then $W^*W = V_s^* V_s + A = I$, and, for every $f \in C(X)$,

$$W^* \widetilde{\rho}(f) W = V_s^* \rho_s(f) V_s + A^{1/2} \pi_\tau(f) A^{1/2} = V_s^* \rho_s(f) V_s + V_y^* \rho_y(f) V_y = \Phi(f).$$

Thus the entire UCP map is reproduced, not only its affine restriction. Give the second target summand the *source* normal label $n(x)$. This is legitimate because both $\pi_\tau(C(X))$ and $A$ commute with $P_n$. Explicitly,

$$(Q_{s,n}(D) \oplus P_n(D))W = W P_n(D).$$

This is the missing normal intertwining. It does not require the normal label to be a function of the nonsmooth target point $\tau(x)$.

*Bounded interpolation on exposed segments.* Use the Borel set $U$ and endpoints $a, b$ from Lemma 5.6. Choose the first coordinate $j$ that varies on each segment, as in that lemma, and partition $U$ into countably many Borel pieces on each of which $j$ is fixed and $b_j(u) - a_j(u) \geq 1/m$ for some fixed positive integer $m$. Such a disjoint partition is obtained by also choosing the least $m$ for which this inequality holds. Since every $F_{n(x)}$, $x \in Z$, is a segment, $P_n(U) = I$.

Write $\widetilde{Q}_n(D) = Q_{s,n}(D) \oplus P_n(D)$. This spectral measure commutes with $\widetilde{\rho}(C(X))$ and satisfies $\widetilde{Q}_n(D)W = W P_n(D)$. In particular $\widetilde{Q}_n(U)W = W$, so its target complement may be discarded. The countable partition then reduces both representations and splits $W$ into isometries between their corresponding summands. Work on one such piece for the next calculation. Simple-function approximation of the projection intertwinings gives

$$c(\widetilde{Q}_n)W = W c(P_n) \qquad \text{for every bounded Borel normal function } c.$$

Here $c(P_n) = \int c(u)\, dP_n(u)$, and similarly on the target.

Fix $f \in C(X)$. On the chosen normal piece set

$$\beta(u) = \frac{f(b(u)) - f(a(u))}{b_j(u) - a_j(u)}, \qquad \alpha(u) = f(a(u)) - \beta(u) a_j(u).$$

Both coefficients are bounded and Borel: the denominator is at least $1/m$, and $K$ is bounded. For either endpoint $t$ of $F_u$, $f(t) = \alpha(u) + \beta(u) t_j$. This identity applies pointwise in all three spectral realizations. The source uses $(u, t) = (n(x), x)$; the smooth target uses $(n(y), y)$ with $y \in S$ and $n(y) \in U$; the second target copy uses $(n(x), \tau(x))$. In each case $t \in X \cap F_u = \{a(u), b(u)\}$. Consequently bounded Borel functional calculus gives

$$\begin{aligned} \widetilde{\rho}(f) &= \alpha(\widetilde{Q}_n) + \beta(\widetilde{Q}_n)\widetilde{\rho}(x_j), \\ \pi(f) &= \alpha(P_n) + \beta(P_n)\pi(x_j). \end{aligned}$$

The normal operators commute with the corresponding representations. The adjoint of the intertwining identity for $\overline{\beta}$ gives $W^* \beta(\widetilde{Q}_n) = \beta(P_n) W^*$, also when $f$ is complex-valued. Using this intertwining and the affine compression identity yields

$$\begin{aligned} W^* \widetilde{\rho}(f) W &= \alpha(P_n) + \beta(P_n) W^* \widetilde{\rho}(x_j) W \\ &= \alpha(P_n) + \beta(P_n)\pi(x_j) = \pi(f). \end{aligned}$$

Combining the countably many normal pieces proves equality on the whole source space. The replacement reproduces $\Phi$, so $\Phi = \pi$. No disintegration is used: all identities are identities of

bounded Borel functions in the original spectral calculi. The lower bound on the segment width is imposed separately on each piece; no uniform lower bound on all segment lengths is assumed. □

*Proof of Theorem 5.1.* After the separable reduction in Lemma 5.2, partition $X$ into $\mathcal{E}$, the countable set $C$, successively disjoint parts of $X \cap F_j$, and the remainder $Z$. All sets are Borel by Lemma 5.5. Lemma 5.7 handles every part except $Z$, and Proposition 5.8 handles $Z$. The direct-sum assertion of Lemma 5.7 concludes the proof. □

The same conclusion holds for every compact convex subset of $\mathbb{R}^3$ with closed extreme boundary: if its affine dimension is at most two, apply the planar theorem after an affine identification with a subset of $\mathbb{R}^2$.

## 6. An explicit counterexample in affine dimension thirty-three

**Theorem 6.1.** *There is an explicitly specified compact convex body $K \subset \mathbb{R}^{33}$, with $\operatorname{aff} K = \mathbb{R}^{33}$ and compact extreme boundary $X = \operatorname{ex} K$, for which the affine function system $A(K)|_X$ is not hyperrigid in $C(X)$. More precisely, there are a unital representation $\pi : C(X) \to B(H)$ and a UCP map $\Psi : C(X) \to B(H)$ such that*

$$\Psi|_{A(K)|_X} = \pi|_{A(K)|_X}, \qquad \Psi(Q) \neq \pi(Q), \qquad Q(x,z) = |x|^2.$$

*The Hilbert space $H$ is separable and infinite dimensional, the spectral measure of $\pi$ has no point atoms, and $\dim_{\mathbb{C}} A(K) = 34$.*

We use the Walsh dilation of Bilich–Scherer [BS, Section 3], with a different three-point distribution. The operator calculation is included in full. Its telescoping identity determines the height of an encoded dilation point. A uniform scalar estimate places all such points in a truncated epigraph with a strongly convex lower graph. An explicit boundary kernel then gives the required UCP map.

### 6.1. The Walsh compression and its height identity.

Fix $R > 1$ and $1 < b < R^2$, and put

$$a = b^{-1}, \qquad \mu = \frac{b}{2R^2}(\delta_{-R} + \delta_R) + \left(1 - \frac{b}{R^2}\right)\delta_0.$$

Thus $\mu$ is symmetric and has second moment $b$. In the discrete alphabet $\{-1, 1, -R, 0, R, *\}$, with a separate star symbol, define

$$\begin{aligned} G &= \{-1,1\}^{\mathbb{N}}, \qquad Y_0 = \{-R,0,R\}^{\mathbb{N}}, \\ Y_k &= \{-1,1\}^{k-1} \times \{*\} \times \{-R,0,R\}^{\{k+1,k+2,\dots\}} \quad (k \geq 1), \\ Y &= G \cup \bigcup_{k \geq 0} Y_k. \end{aligned}$$

The numerical coordinate $x_n$ sends the star to zero. Give $G$ Haar probability $m$. Give $Y_k$ the product probability $\mu_k$ with uniform binary prefix, star at $k$, and $\mu$-distributed tail; on $Y_0$, use $\mu_0 = \mu^{\otimes \mathbb{N}}$. The space $Y$ is compact: unbounded star positions can accumulate only in $G$, and every fixed branch is compact. Each coordinate is continuous, and $Y_0$ is clopen.

Set

$$H = L^2(G, m), \qquad \mathcal{L} = \bigoplus_{k \geq 0} L^2(Y_k, \mu_k).$$

For a finite $F \subset \mathbb{N}$, write $\chi_F = \prod_{n \in F} \varepsilon_n$ and $F_{>k} = F \cap \{k+1, k+2, \dots\}$. For $k \notin F$, the functions

$$\chi_F^{(k)} = b^{-|F_{>k}|/2} \prod_{n \in F} x_n$$

form an orthonormal family in the $k$-th summand. Define on the Walsh basis

$$V\chi_F = a^{|F|/2}\chi_F^{(0)} + \sqrt{1-a}\sum_{k\in F} a^{|F_{>k}|/2}\chi_{F\setminus\{k\}}^{(k)}. \tag{6.1}$$

**Lemma 6.2.** *The map $V : H \to \mathcal{L}$ is an isometry. For bounded Borel functions on $Y$, let $\Phi(f) = V^* M_f V$. Then*

$$\Phi\left(\prod_{n\in E} x_n\right) = M_{\prod_{n\in E}\varepsilon_n} \qquad (E\subset\mathbb{N} \text{ finite}). \tag{6.2}$$

*Also $\Phi(\mathbf{1}_{Y_0})\chi_F = a^{|F|}\chi_F$ and $V1 = \mathbf{1}_{Y_0}$.*

*Proof.* On branch $k$, symmetry and independence make a matrix coefficient of multiplication by $\prod_{n\in E} x_n$ vanish unless the two product labels differ exactly in $E$ and $k \notin E$. In the nonzero case, its value is $b^{|E_{>k}|/2}$: a tail coordinate in the multiplier contributes its second moment divided by $\sqrt{b}$, while common coordinates contribute one after normalization. On branch zero, the same calculation has no star restriction. Consequently, for $C_0 = F \cap F'$,

$$\left\langle M_{\prod_E x_n} V\chi_F, V\chi_{F'}\right\rangle = \mathbf{1}_{\{E=F\triangle F'\}}\left(a^{|C_0|} + (1-a)\sum_{k\in C_0} a^{|(C_0)_{>k}|}\right).$$

The parenthesis is the finite geometric sum $a^r + (1-a)\sum_{j=0}^{r-1} a^j = 1$. Taking $E = \varnothing$ proves isometry on the dense Walsh span; general $E$ proves (6.2). The two final assertions follow by retaining the branch-zero term. □

Let $\beta$ be an even nonnegative function on the numerical digits, with $\beta(\pm1) = 0$. Define diagonal contractions

$$A_n\chi_F = a^{|F_{>n}|}\chi_F \qquad (n \ge 0).$$

The parameters $0 < q < 1$ and $c > 0$ will be chosen so that

$$\begin{aligned} \int (r^2 - \beta(r) - 1)\, d\mu(r) &= c(q^2 - 1), \\ a(R^2 - \beta(R) - 1) + (1-a)(-1 - \beta(0)) &= c(q^2 - a). \end{aligned} \tag{6.3}$$

**Lemma 6.3.** *Under (6.3),*

$$\Phi(x_n^2 - \beta(x_n)) - I = c(q^2 A_n - A_{n-1}) \qquad (n \ge 1). \tag{6.4}$$

*Proof.* Parity makes the left side diagonal. Fix $F$ and put $r = |F_{>n}|$. If $n \notin F$, the squared weights of the branches before $n$ sum to $a^r$; their expected multiplier $x_n^2 - \beta(x_n) - 1$ is the first left side of (6.3). Later branches contribute zero, since their $n$-th coordinate is binary. If $n \in F$, the earlier weights sum to $a^{r+1}$. Under the normalized $x_n^2$-weighted measure, the coordinate has magnitude $R$, so its multiplier is $R^2 - \beta(R) - 1$. The star branch has weight $(1-a)a^r$ and multiplier $-1 - \beta(0)$. The two diagonal coefficients are therefore $a^r$ times the respective left sides of (6.3). These are exactly the coefficients of the right side of (6.4). □

Let $D$ be a positive integer, let $j(n) = 1 + ((n-1) \bmod D)$, and define

$$\begin{aligned} w_n &= q^{n-1} e_{j(n)}, \qquad t(y) = \sum_{n\ge1} w_n x_n(y), \qquad s = t|_G, \\ U(y) &= \sum_{n\ge1} q^{2n-2}\beta(x_n(y)), \\ z_Y(y) &= |t(y)|^2 - U(y) + c\mathbf{1}_{Y_0}(y). \end{aligned} \tag{6.5}$$

These series converge uniformly and absolutely. All the displayed functions are continuous, and $U|_G = 0$. The operator calculation already gives

$$\Phi(t_j) = M_{s_j}, \qquad \Phi(z_Y) = M_{|s|^2}. \tag{6.6}$$

Indeed, all mixed terms in the square expansion of $|t|^2$ compress exactly by (6.2). Termwise compression is valid because $\sum_n |w_n| < \infty$. For the diagonal terms, multiplying (6.4) by $q^{2n-2}$ gives the norm-convergent telescope

$$\sum_{n=1}^{N} q^{2n-2}\left[\Phi(x_n^2 - \beta(x_n)) - I\right] = c(q^{2N}A_N - A_0) \longrightarrow -cA_0.$$

This cancels $c\Phi(\mathbf{1}_{Y_0}) = cA_0$, proving (6.6). No orthogonality or cancellation of distinct vectors $w_n$ is used in this calculation.

6.2. **Numerical choices and uniform containment.** The numerical data are collected below; the other quantities retain their definitions above.

| Quantity | Value | Quantity | Value |
|---|---|---|---|
| $D$ | 32 | $R$ | 12 |
| $b$ | 21/2 | $q^2$ | 45026/72405 |
| $\beta(0)$ | 197/200 | $\beta(R)$ | 2417/20 |
| $c$ | 14481/24320 | Distance coefficient | $1 - 10^{-5}$ |

In particular $\beta(\pm 1) = 0$ completes the specification of $\beta$. The two scalar identities (6.3) are verified exactly: their left sides are respectively

$$-\frac{1441}{6400} = c(q^2 - 1), \qquad \frac{439}{1400} = c(q^2 - a).$$

For the geometric estimate write $\rho = q^D$ and

$$C_\rho = \left\{ \sum_{\ell \geq 0} \rho^\ell \varepsilon_\ell : \varepsilon_\ell \in \{-1, 1\} \right\}.$$

**Lemma 6.4** (Uniform digit estimate)**.** *For every sequence $d_\ell \in \{-R, -1, 0, 1, R\}$,*

$$(1 - 10^{-5}) \operatorname{dist}\left( \sum_{\ell \geq 0} \rho^\ell d_\ell, C_\rho \right)^2 \geq \sum_{\ell \geq 0} \rho^{2\ell} \beta(d_\ell). \tag{6.7}$$

*Proof.* Both sides vanish for an entirely binary sequence. Otherwise let $h$ be its first nonbinary position and put $\delta = \operatorname{dist}(d_h, \{-1, 1\}) \in \{1, R - 1\}$. The largest possible difference between a numerical digit and a binary digit is $R + 1$. Put

$$T = \frac{(R+1)\rho}{1 - \rho}, \qquad E = \frac{\beta(R)\rho^2}{1 - \rho^2}.$$

Against any signed expansion, let $m \leq h$ be the first mismatch. Its leading term dominates the remaining tail and gives a distance at least $\rho^m(|d_m - \varepsilon_m| - T)$. If $m = h$, this is at least $\rho^h(\delta - T)$. If $m < h$, it is at least $\rho^h(R - 1 - T)$, provided $2 - T \geq \rho(R - 1 - T)$. Thus the distance to $C_\rho$ is at least $\rho^h(\delta - T)$. The right side of (6.7) is at most $\rho^{2h}(\beta(d_h) + E)$, since $\beta(R)$ is the largest digit correction.

Here all the required estimates admit rational verification. The integer inequality

$$1990 \cdot 45026^{16} < 72405^{16}$$

gives $\rho < 1/1990$, hence

$$T < \frac{13}{1989} < 1, \qquad E < \frac{2417}{20(1990^2-1)}.$$

Also $2-13/1989 > 11/1990$, which implies the earlier-mismatch comparison $2-T > \rho(R-1-T)$. Finally, clearing positive denominators verifies

$$(1-10^{-5})\left(1-\frac{13}{1989}\right)^2 - \frac{197}{200} - \frac{2417}{20(1990^2-1)} > \frac{19}{10000},$$
$$(1-10^{-5})\left(11-\frac{13}{1989}\right)^2 - \frac{2417}{20} - \frac{2417}{20(1990^2-1)} > \frac{49}{10000}.$$

These prove $(1-10^{-5})(\delta-T)^2 > \beta(d_h)+E$ in both cases. The estimates bound the entire infinite tail. □

6.3. **The convex body and its whole extreme boundary.** The represented horizontal set is the Cartesian product

$$C = s(G) = \prod_{j=1}^{D} q^{j-1} C_\rho. \tag{6.8}$$

Define

$$\begin{aligned} g(x) &= |x|^2 - (1-10^{-5})\operatorname{dist}(x,C)^2 \\ &= 10^{-5}|x|^2 + (1-10^{-5})\sup_{u\in C}(2\langle x,u\rangle - |u|^2). \end{aligned} \tag{6.9}$$

The second formula proves strong convexity: it is a positive quadratic plus a supremum of affine functions. With

$$B = [-R-1, R+1]^D, \qquad M = (R+1)^2 D + c + 1,$$

let

$$K = \{(x,z) : x \in B, \quad g(x) \le z \le M\}. \tag{6.10}$$

This set is compact and convex. Since $g(x) \le |x|^2 \le (R+1)^2 D < M$ throughout $B$, it has nonempty interior in $\mathbb{R}^{D+1}$.

**Lemma 6.5.** *The entire extreme boundary is the compact set*

$$X = \{(x, g(x)) : x \in B\} \cup \{((R+1)\sigma, M) : \sigma \in \{-1,1\}^D\}. \tag{6.11}$$

*Moreover the continuous maps*

$$\iota_G(\varepsilon) = (s(\varepsilon), |s(\varepsilon)|^2), \qquad \iota_Y(y) = (t(y), z_Y(y))$$

*take their values in $X$ and $K$, respectively.*

*Proof.* In a convex decomposition of a lower-graph point, equality in the convexity inequality forces the horizontal coordinates to coincide by strict convexity, and then forces the heights to coincide. Every lower-graph point is therefore extreme. An intermediate-height point is a nontrivial vertical convex combination. The top face is a box, whose only extreme points are its vertices. This proves (6.11), including the absence of any other extreme points. Compactness follows from continuity of $g$ and finiteness of the vertex set.

For the containment assertion, $|t_j(y)| \le Rq^{j-1}/(1-\rho) < R+1$. Also $U \ge 0$, so $z_Y \le |t|^2 + c < M$. Apply Lemma 6.4 to each residue class of digits and multiply by $q^{2(j-1)}$. Squared distance to the product (6.8) is the sum of the squared coordinate distances, so

$$(1-10^{-5})\operatorname{dist}(t(y), C)^2 \ge U(y).$$

Consequently $z_Y(y) \geq g(t(y))$, proving $\iota_Y(Y) \subset K$. Finally $g = |\cdot|^2$ on $C$, so $\iota_G(G) \subset X$. $\square$

6.4. **The boundary kernel and a positive witness.** For $w = (x, z) \in K$, put

$$r(w) = \frac{z - g(x)}{M - g(x)}, \qquad p_\sigma(x) = \prod_{j=1}^{D} \frac{1 + \sigma_j x_j/(R+1)}{2}.$$

The following probability measure on $X$ has barycenter $w$:

$$\nu_w = (1 - r(w))\delta_{(x,g(x))} + r(w) \sum_{\sigma \in \{-1,1\}^D} p_\sigma(x)\delta_{((R+1)\sigma, M)}. \tag{6.12}$$

Indeed all weights are nonnegative and sum to one, and $\sum_\sigma p_\sigma(x)(R+1)\sigma = x$. The denominator $M - g(x)$ is positive on the whole box. For every $f \in C(X)$, the function $w \mapsto \int f \, d\nu_w$ is continuous.

Define

$$\pi(f) = M_{f \circ \iota_G}, \qquad \widehat{f}(y) = \int_X f \, d\nu_{\iota_Y(y)}, \qquad \Psi(f) = V^* M_{\widehat{f}} V. \tag{6.13}$$

The first map is a unital representation. Pointwise integration against probability measures preserves positivity at every matrix level; multiplication and isometric compression therefore make $\Psi$ UCP. This formula uses the explicit finite kernel (6.12), with no choice of representing measures.

It also gives a representation dilation directly. Index the terms of (6.12) by $J = \{0\} \cup \{-1, 1\}^D$, writing their locations and weights at $\iota_Y(y)$ as $F_\ell(y) \in X$ and $\omega_\ell(y) \geq 0$. On $\widetilde{\mathcal{L}} = \bigoplus_{k \geq 0} \bigoplus_{\ell \in J} L^2(Y_k, \mu_k)$, let $\rho_X(f)$ multiply component $(k, \ell)$ by $f(F_\ell(y))$. The map

$$(W\xi)_{k,\ell}(y) = \sqrt{\omega_\ell(y)}\, \xi_k(y)$$

is an isometry, and $\Psi(f) = (WV)^* \rho_X(f)(WV)$.

*Completion of the proof of Theorem 6.1.* For an affine function, its integral against $\nu_w$ is its value at $w$. Thus (6.6) gives

$$\Psi(x_j|_X) = M_{s_j} = \pi(x_j|_X), \qquad \Psi(z|_X) = M_{|s|^2} = \pi(z|_X).$$

Together with constants, these coordinates span $A(K)|_X$. They are linearly independent because $K$ has full affine dimension $D + 1 = 33$, and restriction to $X$ is injective because $K = \operatorname{conv} X$. Hence the function system has complex dimension $D + 2 = 34$.

For $Q(x, z) = |x|^2$, scalar convexity and the barycenter property give $\widehat{Q}(y) \geq |t(y)|^2$. The constant Walsh vector has norm one and $V1 = \mathbf{1}_{Y_0}$. Independence and zero means remove all mixed terms from the following scalar integrals:

$$\langle \Psi(Q)1, 1\rangle \geq \int_{Y_0} |t|^2 \, d\mu_0 = \frac{b}{1 - q^2},$$
$$\langle \pi(Q)1, 1\rangle = \int_G |s|^2 \, dm = \frac{1}{1 - q^2}.$$

Consequently

$$\langle [\Psi(Q) - \pi(Q)]1, 1\rangle \geq \frac{b - 1}{1 - q^2} > 0.$$

The representation therefore fails the unique extension property.

Finally $q^D = \rho < 1/2$ makes each signed expansion in $C_\rho$ injective: the first differing digit exceeds the entire remaining difference tail. The product encoding $s$, and hence $\iota_G$, is injective. A point inverse image under $\iota_G$ is therefore empty or a singleton; Haar measure on $G$ assigns it

measure zero. This proves that the spectral measure of $\pi$ has no point atoms. The countably infinite Walsh basis proves that $H$ is separable and infinite dimensional. □

## 7. An obstruction for stationary geometric Walsh encodings

The construction above separates the operator identities from the geometric containment problem. We now show that its dimension cannot be reduced to four by changing only the fixed three-point distribution, the geometric decay rate, or the directions of the encoding vectors. The first obstruction concerns the quadratic distance envelope. The next section extends it, in affine dimension four, to every finite strictly convex lower graph.

### 7.1. **A signing estimate.**

**Lemma 7.1** (Signed series in finite dimension)**.** *Let $w_n \in \mathbb{R}^D$ satisfy $\sum_n |w_n| < \infty$, and let $L_D$ be the sum of the $D$ largest numbers among $\{|w_n|^2 : n \geq 1\}$, padding with zeros if necessary. There are signs $\varepsilon_n \in \{-1, 1\}$ such that*

$$\left|\sum_{n\geq 1} \varepsilon_n w_n\right|^2 \leq L_D.$$

*Proof.* For a finite sequence $w_1, \ldots, w_N$, the polytope

$$\left\{a \in [-1,1]^N : \sum_{n=1}^{N} a_n w_n = 0\right\}$$

is nonempty. At an extreme point $a$, at most $D$ coordinates lie strictly between $-1$ and 1: otherwise the corresponding vectors are linearly dependent and admit a nonzero feasible perturbation in both directions. Independently round each of these coordinates to a sign with mean $a_n$, leaving the other coordinates fixed. Since the centered rounding errors are independent,

$$\mathbb{E}\left|\sum_{n=1}^{N} \varepsilon_n w_n\right|^2 = \sum_{|a_n|<1} (1 - a_n^2)|w_n|^2 \leq L_D.$$

Some realization therefore has the required bound. Apply this argument to finite prefixes and extend their signings arbitrarily. The remaining tail has norm at most $\sum_{n>N} |w_n| \to 0$. Compactness of the sign space and uniform convergence of the signed series give the assertion for the infinite sequence. □

### 7.2. **Eliminating all fixed three-point parameters.**

In this section, $\gamma$ denotes the squared geometric ratio; thus $\gamma = q^2$ in the construction above. The numerical tail law is supported on $\{0, \pm R\}$, with variance $b$ and masses $1 - b/R^2, b/(2R^2), b/(2R^2)$. As before, $G$ is the binary source, $Y_0$ has only tail coordinates, and $Y_k$ has a binary prefix, a zero-valued star at $k$, and tail coordinates thereafter.

**Theorem 7.2** (No stationary geometric distance encoding in horizontal dimension at most four)**.** *Let*

$$R > 1, \qquad 1 < b < R^2, \qquad c > 0, \qquad 0 < \gamma < 1, \qquad u, v \geq 0.$$

*Suppose the exact height equations hold:*

$$\begin{aligned} \frac{b}{R^2}(R^2 - 1 - v) + \left(1 - \frac{b}{R^2}\right)(-1 - u) &= c(\gamma - 1), \\ \frac{1}{b}(R^2 - 1 - v) + \left(1 - \frac{1}{b}\right)(-1 - u) &= c(\gamma - 1/b). \end{aligned} \tag{7.1}$$

*Let $w_n \in \mathbb{R}^D$ have arbitrary directions and satisfy $|w_n|^2 = \gamma^{n-1}$. Define*

$$t(y) = \sum_{n\geq 1} w_n x_n(y), \qquad C = t(G), \qquad U(y) = \sum_{n\geq 1} \gamma^{n-1}\beta(x_n(y)),$$

*where $\beta(\pm 1) = 0$, $\beta(0) = u$, and $\beta(\pm R) = v$. If $D \leq 4$, there is no $0 < \lambda \leq 1$ for which*

$$\lambda \operatorname{dist}(t(y), C)^2 - U(y) + c\mathbf{1}_{Y_0}(y) \geq 0 \qquad (y \in Y). \tag{7.2}$$

*Proof.* Two consequences of Lemma 7.1 suffice. At a star in the first position followed by zero digits, the encoded point is zero and $U = u/(1-\gamma)$. Signing the source series bounds its distance to $C$. On $Y_0$, take digits $R\varepsilon_n$ and compare with the source point having digits $\varepsilon_n$; the difference is $(R-1)\sum_n \varepsilon_n w_n$. Thus (7.2) implies, after discarding $\lambda \leq 1$,

$$u \leq 1 - \gamma^D, \qquad v - c(1-\gamma) \leq (R-1)^2(1-\gamma^D). \tag{7.3}$$

For $D \leq 4$ these inequalities remain necessary with $D$ replaced by four.

Put $x = \gamma^4$, $h = 1 + u$, and $\Delta = R^2 + u - v$. The height equations give

$$c = \frac{\Delta(R^2 - b^2)}{R^2(b-1)}, \qquad c(1-\gamma) = h - \frac{b}{R^2}\Delta. \tag{7.4}$$

Using the second equality in the second inequality of (7.3) yields

$$\left(1 - \frac{b}{R^2}\right)\Delta \geq 2(R-1) + (R-1)^2 x > 0. \tag{7.5}$$

Hence $\Delta > 0$, and the first equality of (7.4), together with $c > 0$, forces $b < R$. In particular all subsequent denominators are positive. Eliminating $c$ also gives

$$\frac{h}{\Delta} = \frac{R^2 - b - \gamma(R^2 - b^2)}{R^2(b-1)} > 0.$$

Multiply (7.5) by $h/\Delta$ and use $h \leq 2 - x$, from the first inequality in (7.3). After cancellation,

$$(R-1)(2 + (R-1)x)\left(1 - \gamma\frac{R^2 - b^2}{R^2 - b}\right) \leq (b-1)(2-x). \tag{7.6}$$

Set $r = R - 1 > 0$, $\alpha = (b-1)/(R-1) \in (0,1)$, and $A = 1 - \alpha \in (0,1)$. Clearing the positive denominators in (7.6) says that $p(r) \geq 0$, where

$$p(r) = \gamma(1-\alpha)(2 + (1+\alpha)r)(2 + rx) - (r + 2 - \alpha)(2(1-\alpha) + (r+\alpha)x).$$

Its quadratic, linear, and constant coefficients are respectively

$$\begin{aligned} &x(\gamma(1-\alpha^2) - 1), \\ &2\big[-\gamma A^2 + (2\gamma - 1 + \gamma x)A - x\big], \\ &-(2-x)A^2 + 2(2\gamma - 1)A - x. \end{aligned} \tag{7.7}$$

We prove that all three are strictly negative. The first is immediate. For the second put $N = 2\gamma - 1 + \gamma^5$. If $N \leq 0$, negativity is immediate; if $N > 0$, half the coefficient is at most $N^2/(4\gamma) - \gamma^4$. We claim that

$$2\gamma - 1 + \gamma^5 < 2\gamma^{5/2} \qquad (0 < \gamma < 1). \tag{7.8}$$

Indeed, setting $t = \sqrt{\gamma}$, the difference factors as

$$t^{10} - 2t^5 + 2t^2 - 1 = (t-1)Q(t),$$

where

$$Q(t) = t^9 + t^8 + t^7 + t^6 + t^5 - t^4 - t^3 - t^2 + t + 1 > 0 \qquad (0 < t < 1).$$

For $t \le 2/3$, this follows from $t^2+t^3+t^4 \le 76/81 < 1$. For $t \ge 2/3$, use

$$Q(t) = (1-t^2)(1+t) + t^4(t+t^2+t^3+t^4+t^5-1),$$

whose first term is positive and whose second is nonnegative because $t+t^2 \ge 10/9$. This proves (7.8) and negativity of the linear coefficient.

For the constant coefficient in (7.7), $\gamma \le 1/2$ is immediate. If $\gamma > 1/2$, its maximum over real $A$ is

$$\frac{(2\gamma-1)^2}{2-\gamma^4} - \gamma^4 < 0.$$

To verify the strict inequality, factor its numerator as

$$(2\gamma-1)^2 - 2\gamma^4 + \gamma^8 = (\gamma-1)(\gamma^7+\gamma^6+\gamma^5+\gamma^4 - \gamma^3-\gamma^2+3\gamma-1).$$

The parenthesis is positive, since $3\gamma-1-\gamma^2-\gamma^3 = (1-\gamma)(\gamma^2+2\gamma-1) \ge 0$ on this interval and the remaining powers are positive. All coefficients of $p$ are therefore negative, so $p(r) < 0$ for $r > 0$, a contradiction. □

Theorem 7.2 permits arbitrary, nonperiodic directions and every fixed symmetric three-point distribution with nonnegative potential satisfying the stated height equations. It excludes the distance-envelope realization in affine dimensions four and five, but is not a universal lower bound for commutative counterexamples. The remaining possibility of changing the lower graph requires an operator argument.

## 8. Strictly convex graphs in affine dimension four

The quadratic envelope is forced by an injective affine defect when the horizontal dimension is at most three. The proof requires no smoothness of the lower graph. Away from its quadratic tangencies, supporting affine functions admit a metric taking values on a positive distance level. Such levels have the critical covering bound needed by Theorem 3.1.

### 8.1. **Distance levels and supporting affine functions.**

**Lemma 8.1** (Covering a positive distance level)**.** *Let $C \subset \mathbb{R}^D$ be nonempty and compact, and fix $\varepsilon > 0$. The compact set*

$$S_\varepsilon = \{z : \operatorname{dist}(z, C) = \varepsilon\}$$

*has covers by $O(r^{-(D-1)})$ sets of diameter at most $r$ as $r \downarrow 0$.*

*Proof.* For $z \in S_\varepsilon$, choose a nearest point $c \in C$ and put $v = (z-c)/\varepsilon$. If $z' \in S_\varepsilon$ and $d = z'-z$, then $|z'-c| \ge \varepsilon$. Interchanging the two points gives, for any nearest-point unit vector $v'$ at $z'$,

$$v \cdot d \ge -\frac{|d|^2}{2\varepsilon}, \qquad v' \cdot d \le \frac{|d|^2}{2\varepsilon}. \tag{8.1}$$

Cover $S_\varepsilon$ by finitely many closed spatial sets of diameter at most $\varepsilon/4$, and the unit sphere by finitely many closed balls of radius $1/8$ with unit centers $e_j$. Partition into overlapping patches by requiring membership in one spatial set and the existence of a nearest-point vector within $1/8$ of one $e_j$. These patches are compact, since they are projections of compact sets of nearest-point pairs.

For two points in the same patch, choose witnessing vectors in (8.1). Then

$$|e_j \cdot d| \le \frac{1}{8}|d| + \frac{|d|^2}{2\varepsilon} \le \frac{1}{4}|d|.$$

Orthogonal projection onto $e_j^\perp$ is therefore injective on the patch and has a Lipschitz inverse. A grid in its bounded projected subset of $\mathbb{R}^{D-1}$ gives the stated covering bound. Sum over the finitely many patches. For $D = 1$, each patch contains at most one point. No continuity or uniqueness of nearest-point vectors is needed. □

For a finite convex function $g$ on an open convex set $\Omega$, we write

$$\partial g(s) = \{p : \ g(t) \geq g(s) + \langle p, t - s\rangle \text{ for all } t \in \Omega\}$$

for its convex subdifferential. The next lemma uses subgradients that differ from the gradient $2s$ of the quadratic source graph.

**Lemma 8.2** (A metric for subgradient barriers)**.** *Let $g$ be finite and convex on an open convex set $\Omega \subset \mathbb{R}^D$, and let $C \subset \Omega$ be compact with $g(s) = |s|^2$ on $C$. Suppose $F \subset C$ is Borel and $p_s \in \partial g(s)$ is a Borel selection satisfying $0 < \delta \leq |p_s - 2s| \leq M < \infty$ on $F$. Set*

$$v_s = \frac{p_s - 2s}{|p_s - 2s|}, \qquad \varepsilon = \delta/16, \qquad \theta_s = s + \varepsilon v_s.$$

*Then $\theta : F \to S_\varepsilon$ is Borel, with the distance level taken relative to the whole set $C$, and for a finite constant $L$ depending only on $\delta, M$,*

$$0 \leq |t|^2 - |s|^2 - p_s \cdot (t - s) \leq L|\theta_t - \theta_s|^2 \qquad (s, t \in F). \tag{8.2}$$

*Proof.* The subgradient inequality and $g = |\cdot|^2$ on $C$ give

$$(p_s - 2s) \cdot (t - s) \leq |t - s|^2 \qquad (s \in F, \ t \in C).$$

Consequently

$$|\theta_s - t|^2 \geq \varepsilon^2 + (1 - 2\varepsilon/\delta)|t - s|^2.$$

Thus $s$ is the unique nearest point in $C$ to $\theta_s$, and $\theta_s \in S_\varepsilon$.

For $s, t \in F$ write $d = t - s$ and $w = v_t - v_s$. Applying the subgradient inequality in both orders gives

$$v_s \cdot d \leq |d|^2/\delta, \qquad v_t \cdot d \geq -|d|^2/\delta,$$

so $w \cdot d \geq -2|d|^2/\delta$ and

$$|\theta_t - \theta_s|^2 = |d + \varepsilon w|^2 \geq \tfrac{3}{4}|d|^2 + \varepsilon^2|w|^2. \tag{8.3}$$

The barrier in (8.2) is nonnegative by convexity. Writing $\alpha_s = |p_s - 2s| \leq M$, its upper bound is

$$|d|^2 - \alpha_s v_s \cdot d \leq (1 + M/\delta)|d|^2 + M|w|\,|d|.$$

Young's inequality and (8.3) bound this by $L|\theta_t - \theta_s|^2$ for a finite $L$. □

## 8.2. **An injective defect forces the quadratic envelope.**

**Theorem 8.3** (An injective defect forces the quadratic envelope)**.** *Let $1 \leq D \leq 3$, and let $g$ be finite and strictly convex on an open convex set $\Omega \subset \mathbb{R}^D$. Let $K \subset \Omega \times \mathbb{R}$ be compact and convex, with $X = \operatorname{ex} K$ compact and $K \subset \{(t, z) : z \geq g(t)\}$. Suppose $C \subset \Omega$ is nonempty and compact, and*

$$g(s) = |s|^2, \qquad x_s = (s, |s|^2) \in K \qquad (s \in C).$$

*The graph points form a set $E = \{x_s : s \in C\} \subset X$. Let $\pi : C(X) \to B(H)$ be a unital representation concentrated on $E$ and having full spectral support there. Suppose a UCP map $\Psi : C(X) \to B(H)$ agrees with $\pi$ on affine functions, and for some real affine $h$ the operator*

$$\mathcal{D}_h = \Psi(h^2) - \pi(h)^2$$

*is injective. Then $\pi$ is concentrated on points where $\partial g(s) = \{2s\}$, equivalently where $g$ is differentiable with $\nabla g(s) = 2s$. Moreover,*

$$2s \in \partial g(s) \quad (s \in C), \qquad g(t) \geq |t|^2 - \operatorname{dist}(t, C)^2 \quad (t \in \Omega). \tag{8.4}$$

*Proof.* Subdifferentials of a finite convex function on an open convex domain are nonempty and compact, locally bounded, and have closed graph. Local boundedness follows by evaluating the supporting inequality a fixed small distance in every direction inside a larger compact ball. Hence the subgradient graph over $C$ is compact, and $|p - 2s| \leq M$ there for some finite $M$.

For $m \geq 1$ let

$$\Gamma_m = \{(s, p) : s \in C,\ p \in \partial g(s),\ |p - 2s| \geq 1/m\}, \qquad F_m = \operatorname{pr}_C \Gamma_m.$$

These sets are compact. The increasing sets $F_m$ exhaust $\{s \in C : \partial g(s) \neq \{2s\}\}$. On each nonempty $F_m$, Lemma 4.3 gives a Borel selector with $1/m \leq |p_s - 2s| \leq M$. For the graph $E_m = \{x_s : s \in F_m\}$ define

$$b_{x_s}(t, z) = z - |s|^2 - p_s \cdot (t - s).$$

The coefficients are Borel, and $b_{x_s}$ is nonnegative on $K$ and vanishes there only at $x_s$. Indeed strict convexity makes the subgradient inequality strict at every $t \neq s$; at $t = s$, equality requires $z = g(s)$. This also shows that all graph points under consideration are exposed extreme points of $K$.

Lemma 8.2, with $\delta = 1/m$, bounds these barriers on $E_m$ by a constant times squared distance after a Borel map into $S_{1/(16m)}$. By Lemma 8.1, this compact level has $O(r^{-(D-1)})$ covers and hence, since $D \leq 3$, $O(r^{-2})$ covers. Lemma 3.3 gives infinite inverse-square energy for every nonzero finite measure pushed onto it. Theorem 3.1 therefore gives UEP for every representation concentrated on $E_m$.

Write $P$ for the spectral measure of $\pi$ and $P_m = P(E_m)$. Restrict $\pi$ and compress $\Psi$ to each nonzero $P_m H$. These maps still agree on the affine functions, so the preceding UEP assertion makes them equal. Since $P_m$ commutes with $\pi(h)$,

$$P_m \mathcal{D}_h P_m = 0.$$

The operator $\mathcal{D}_h$ is positive by the Schwarz inequality. Thus $\mathcal{D}_h^{1/2} P_m = 0$, and injectivity forces $P_m = 0$. The Borel set

$$E_{\mathrm{bad}} = \bigcup_m E_m = \{x_s : s \in C,\ \partial g(s) \neq \{2s\}\}$$

has spectral projection zero. A finite convex function is differentiable precisely where its subdifferential is a singleton, which proves the concentration assertion.

Full spectral support makes the remaining source points dense in $C$. The closed subgradient graph then gives $2s \in \partial g(s)$ for every $s \in C$. Finally,

$$g(t) \geq \sup_{s \in C} (g(s) + 2s \cdot (t - s)) = |t|^2 - \operatorname{dist}(t, C)^2,$$

proving (8.4). □

**Corollary 8.4** (UEP away from quadratic tangencies)**.** *Under the geometric hypotheses of Theorem 8.3, every representation concentrated on $E_{\mathrm{bad}} = \{x_s : s \in C,\ \partial g(s) \neq \{2s\}\}$ has UEP. More generally, for any representation $\pi$ concentrated on $E$ and any affine-agreeing UCP map $\Psi$,*

$$(\Psi(h^2) - \pi(h)^2) P(E_{\mathrm{bad}}) = 0$$

*for every real affine $h$.*

*Proof.* Use the increasing compact sets $E_m$ in the preceding proof, each of which supports only representations with UEP. If $P(E_{\mathrm{bad}}) = I$, then $P(E_m) \uparrow I$ and UEP on the compressed spaces gives

$$P(E_m)\Psi(f)P(E_m) = P(E_m)\pi(f)P(E_m) \qquad (f \in C(X)).$$

Taking strong limits proves $\Psi = \pi$. Without that concentration assumption, compressed equality still gives $P(E_m)\mathcal{D}_h P(E_m) = 0$. Positivity yields $\mathcal{D}_h P(E_m) = 0$, and the strong limit gives the result. □

8.3. **Application to the Walsh construction.**

**Lemma 8.5** (Injective horizontal defects)**.** *For the stationary three-point Walsh compression, assume $1 < b < R^2$ and write $\Phi(f) = V^* M_f V$. Each operator $D_n = \Phi(x_n^2) - I$ is positive and injective. If $t_j = \sum_n (w_n)_j x_n$ is a nonzero linear coordinate with $\sum_n R|w_n| < \infty$ and $s_j = t_j|_G$, then*

$$\Phi(t_j^2) - M_{s_j}^2 = \sum_n (w_n)_j^2 D_n \tag{8.5}$$

*is also positive and injective. A positive barycentric-kernel realization with horizontal coordinate $t_j$ has an actual coordinate defect at least as large as this operator.*

*Proof.* Put $a = 1/b$. On the Walsh vector $\chi_F$, let $r = |F_{>n}|$. If $n \notin F$, the total squared weight of branches before $n$ is $a^r$, and their tail second moment is $b$; later branches have binary value squared equal to one. If $n \in F$, the total weight before $n$ is $a^{r+1}$, their squarefree-product-weighted second moment is $R^2$, and branch $n$ has weight $(1-a)a^r$ and value zero. Consequently

$$D_n \chi_F = a^r \begin{cases} b - 1, & n \notin F, \\ R^2/b - 1, & n \in F. \end{cases} \chi_F.$$

Every diagonal entry is strictly positive, so $D_n$ is injective; no uniform positive lower bound is asserted.

Uniform absolute convergence permits expansion of $t_j^2$. All mixed terms compress exactly by (6.2), leaving the norm-convergent positive sum (8.5). One nonzero coefficient bounds this sum below by a positive scalar multiple of an injective $D_n$. Finally, scalar Jensen for a probability measure with barycenter $(t(y), z(y))$ gives $\int \tau_j^2 \, d\nu_y \geq t_j(y)^2$. Applying the positive compression shows $\Psi(\tau_j^2) \geq \Phi(t_j^2)$, while affine agreement gives $\pi(\tau_j) = M_{s_j}$. This proves the last assertion. □

**Corollary 8.6** (No strictly convex graph in affine dimension four)**.** *Take the stationary geometric data of Theorem 7.2. Suppose a compact convex body $K$ with compact extreme boundary contains every encoded point*

$$(t(y), z_Y(y)), \qquad z_Y(y) = |t(y)|^2 - U(y) + c\mathbf{1}_{Y_0}(y), \qquad y \in Y,$$

*and lies in the epigraph of a finite strictly convex function $g$ on an open convex horizontal domain containing $t(Y)$. Assume $g(s) = |s|^2$ on $C = t(G)$, the signed graph is represented on $L^2(G, m)$, and a positive barycentric kernel realizes an affine-agreeing UCP map. Then $D \geq 4$. In particular this mechanism cannot give an affine-dimension-four counterexample, with or without differentiability of $g$.*

*Proof.* The continuous surjection $t : G \to C$ sends Haar measure to a measure of full support: the preimage of a nonempty relatively open set is nonempty and open, and therefore has positive Haar measure. This does not require injectivity of the encoding. Some horizontal coordinate has a nonzero coefficient, and Lemma 8.5 gives an injective defect for the actual barycentric map.

If $D \le 3$, Theorem 8.3 gives $g(t) \ge |t|^2 - \operatorname{dist}(t, C)^2$. Containment of the encoded points therefore implies

$$\operatorname{dist}(t(y), C)^2 - U(y) + c\mathbf{1}_{Y_0}(y) \ge 0 \qquad (y \in Y),$$

contradicting Theorem 7.2 with $\lambda = 1$. □

*Remark* 8.7. Theorem 8.3 itself does not assume a Walsh model, stationarity, or geometric vector norms. It uses one injective affine defect and full spectral support on the quadratic graph. Its dimension restriction is $D - 1 \le 2$, the covering exponent of the auxiliary distance level. Accordingly, the strictly convex graph obstruction applies in affine dimension four; the stronger affine-dimension-five exclusion in Theorem 7.2 applies to the distance-envelope certificate. These are construction-family obstructions, not universal dimension lower bounds. They leave open constructions using a different compression, nongeometric vector norms, other height identities, or geometry outside the strictly convex graph setting.

## 9. A rank-one defect cannot produce a counterexample

**Theorem 9.1** (Rank-one total quadratic defects are impossible)**.** *Let $K \subset \mathbb{R}^d$ be a nonempty compact convex set whose extreme boundary $X = \operatorname{ex} K$ is compact. Let $\pi : C(X) \to B(H)$ be a unital $*$-representation and let $\Psi : C(X) \to B(H)$ be a unital completely positive map agreeing with $\pi$ on the restrictions of affine functions on $K$. Write $x_1, \dots, x_d$ for the coordinate functions and set*

$$D = \sum_{j=1}^{d} (\Psi(x_j^2) - \pi(x_j)^2).$$

*If* $\operatorname{rank} D \le 1$*, then $\Psi = \pi$. In particular, $D$ cannot have rank one.*

*Proof.* Take a Stinespring dilation and identify its isometric copy of $H$ with a subspace of the dilation space $H \oplus E$. Write

$$N_j = \rho(x_j) = \begin{pmatrix} A_j & B_j \\ B_j^* & C_j \end{pmatrix}, \qquad A_j = \pi(x_j).$$

The $N_j$ commute, the $A_j$ commute, and

$$\Psi(x_j^2) - A_j^2 = B_j B_j^*, \qquad D = \sum_j B_j B_j^*.$$

Suppose $\operatorname{rank} D = 1$ and let $h$ be a unit vector spanning its range. There are vectors $k_j \in E$ such that $B_j = hk_j^*$, with at least one $k_j \ne 0$. Put $Q = I_H - hh^*$ and $v_j = QA_jh$. Applying $Q$ to the upper-right block of $N_iN_j = N_jN_i$ gives

$$v_i k_j^* = v_j k_i^*. \tag{9.1}$$

If $\operatorname{span}_{\mathbb{C}}\{k_1, \dots, k_d\}$ has dimension at least two, choose independent $k_i, k_j$. Equation (9.1) implies $v_i = v_j = 0$: evaluate it on a vector orthogonal to one of $k_i, k_j$ but not the other. Pairing any remaining index with $i$ then gives $v_\ell = 0$. Thus $h$ is a joint eigenvector of the $A_j$.

If that span has dimension one, the upper-left commutation equations give $B_iB_j^* = B_jB_i^*$. Choosing a nonzero $B_1$, after relabeling, it follows that $B_j = \lambda_j B_1$ with $\lambda_j \in \mathbb{R}$ and $\lambda_1 = 1$. Equation (9.1) therefore yields

$$(A_j - \lambda_j A_1)h = c_j h, \qquad c_j \in \mathbb{R}.$$

The joint spectral measure of $h$ is supported on the affine line defined by $x_j - \lambda_j x_1 = c_j$ for $j = 2, \dots, d$. An affine line meets $\operatorname{ex} K$ in at most two points. Consequently, in both cases $h$ lies in a sum of at most two joint eigenspaces.

Let $e \neq 0$ be a joint eigenvector at $x \in X$. The scalar state

$$f \longmapsto \frac{\langle \Psi(f)e, e\rangle}{\|e\|^2}$$

has barycenter $x$. Since $x$ is extreme, its representing probability measure is $\delta_x$. Hence

$$\|B_j^* e\|^2 = \langle (\Psi(x_j^2) - A_j^2)e, e\rangle = 0 \qquad (1 \le j \le d).$$

Apply this to the at most two spectral components of $h$. It follows that $B_j^* h = 0$ for every $j$, contradicting $B_j^* h = k_j$ and the existence of a nonzero $k_j$. This excludes rank one.

If $D = 0$, every positive summand $B_j B_j^*$ is zero. Thus all coordinates belong to the multiplicative domain of $\Psi$. They generate $C(X)$, so $\Psi = \pi$. □

The theorem concerns the total quadratic defect, rather than Kraus rank. Its rank is unchanged by an invertible real affine change of coordinates: the closure of its range is the closed span of the ranges of the $B_j$. No conclusion for defects of rank two or higher is asserted here.

## 10. A flat-family dimension bound

**Lemma 10.1.** *Let $E$ be a complex Hilbert space, and let $\mathcal{U}$ be a finite-dimensional real linear subspace of $B(\mathbb{C}^m, E)$, with $m < \infty$. If*

$$AB^* = BA^* \qquad (A, B \in \mathcal{U}),$$

*then* $\dim_{\mathbb{R}} \mathcal{U} \le m$.

*Proof.* For $x \in E$ define the real-linear map $L_x : \mathcal{U} \to \mathbb{C}^m$ by $L_x(A) = A^* x$. Choose $x$ maximizing its real rank $r$. We claim that

$$L_y(\ker L_x) \subset \operatorname{ran}_{\mathbb{R}} L_x \qquad (y \in E).$$

Indeed, choose real domain and range decompositions in which $L_x$ has an invertible $r \times r$ upper-left block and zero remaining blocks. If the lower-right block of $L_y$ were nonzero, an $(r+1) \times (r+1)$ minor of $L_x + tL_y$ would have a nonzero linear term in real $t$. Its rank would exceed $r$ for small $t \neq 0$, contradicting $L_x + tL_y = L_{x+ty}$ and the choice of $x$.

Now suppose $A^* x = 0$. For every $B \in \mathcal{U}$, $AB^* x = BA^* x = 0$. Hence $A^* y$ is complex-orthogonal to every $B^* x$, for every $y \in E$. The claimed inclusion puts $A^* y$ in the real span of those same vectors. Therefore $A^* y = 0$ for every $y$, so $A = 0$. Thus $L_x$ is injective.

The real subspace $\operatorname{ran}_{\mathbb{R}} L_x \subset \mathbb{C}^m$ is isotropic for $(v, w) \mapsto \operatorname{Im}\langle v, w\rangle$: the cross-product hypothesis makes $AB^*$ self-adjoint, so $\langle A^* x, B^* x\rangle$ is real. This subspace is real-orthogonal to its product by $i$; its real dimension is at most $m$. Injectivity concludes the proof. □

This resembles the flat-bilinear-form identities from submanifold geometry, but the proof requires no classification theorem and handles rank-deficient families.

## 11. Block Hardy compressions

Let $T_F = P_+ M_F|_{H^2 \otimes \mathbb{C}^m}$ for a bounded measurable $m \times m$ matrix symbol $F$. Let $S$ be the multiplicity-$m$ unilateral shift and $E_0 : \mathbb{C}^m \to H^2 \otimes \mathbb{C}^m$ the constant-function inclusion. We use $H_G = (I - P_+) M_G|_{H^2 \otimes \mathbb{C}^m}$ for the Hankel operator.

**Proposition 11.1.** *A finite commuting self-adjoint block Toeplitz tuple of multiplicity $m$ admits a finite orthogonal decomposition by constant coefficient spaces*

$$\mathbb{C}^m = \bigoplus_{\alpha} E_\alpha$$

*such that on each block both its joint spectrum and the multiplication-symbol joint spectrum lie in the same affine subspace of dimension at most* $\dim_{\mathbb{C}} E_\alpha$. *The coefficient dimensions sum to* $m$.

*Proof.* For completeness, commuting Toeplitz operators have commuting symbols. The semi-commutator identity $T_{FG} - T_F T_G = H^*_{F^*} H_G$ and $H_G S^n \to 0$ strongly imply $S^{*n} T_F T_G S^n \to T_{FG}$ strongly. Here the convergence of the Hankel tails follows first on analytic polynomials from square summability of Fourier coefficients and then by boundedness. Thus $[T_F, T_G] = 0$ implies $T_{FG-GF} = 0$, hence $FG = GF$ almost everywhere, by injectivity of the Toeplitz-symbol map. Self-adjoint Toeplitz operators likewise have Hermitian symbols.

Write $\mathcal{W}$ for the real span of the identity and the tuple. For $A \in \mathcal{W}$ put $U_A = S^* A E_0$. Since $S^* A S = A$,

$$S^*[A, B]S - [A, B] = U_A U_B^* - U_B U_A^*.$$

Lemma 10.1 bounds the real dimension of $\{U_A : A \in \mathcal{W}\}$ by $m$. The kernel of $A \mapsto U_A$ consists exactly of constant Hermitian matrices $I \otimes C$, since $U_A$ records every positive Fourier coefficient and self-adjointness determines the negative coefficients.

If this kernel consists only of scalar constants, then $\dim_{\mathbb{R}} \mathcal{W} \le m + 1$. The coordinate joint spectrum therefore lies in an affine subspace of dimension at most $m$. Every affine relation is also a relation between the symbols, by Toeplitz injectivity, so their joint spectrum lies in the same subspace.

Otherwise choose a nonscalar constant Hermitian matrix in the kernel. It commutes with the whole tuple. Its eigenspaces reduce both the Toeplitz operators and their symbols: the corresponding symbol commutators have zero Toeplitz operator and hence vanish. Split into these proper coefficient subspaces and repeat the argument on each restricted tuple. Each split decreases the relevant positive integer coefficient dimension, so the procedure terminates in blocks with only scalar constants in their kernels. The previous paragraph then applies to each block. □

**Corollary 11.2.** *Let* $K \subset \mathbb{R}^d$ *be compact convex with* $X = \operatorname{ex} K$ *compact, and let* $\rho : C(X) \to L^\infty(\mathbb{T}, M_m)$ *be a unital representation. Suppose compression to* $H^2 \otimes \mathbb{C}^m$ *agrees on the affine coordinates with a representation* $\pi$ *of* $C(X)$. *If* $m \le 3$, *the compression is* $\pi$ *on all of* $C(X)$.

*Proof.* Apply Proposition 11.1 to the compressed coordinate tuple. The constant coefficient blocks reduce the full representations, because the coordinates generate $C(X)$. On each block let $L$ be the common affine subspace. Both representations factor through $C(Y)$, where $Y = X \cap L$ is compact. Moreover $Y = \operatorname{ex}(\operatorname{conv} Y)$: each point of $Y$ remains extreme in the smaller convex set, and extreme points of the convex hull of a compact set belong to that set. Ambient affine functions restrict onto all affine functions of this convex hull.

For $m \le 2$, the planar theorem applies on every block. For $m \le 3$, Theorem 5.1 covers the only additional case, after identifying the affine span with $\mathbb{R}^3$. Each compression is therefore the source representation, and so is their finite sum. □

*Remark* 11.3. The target-symbol spectrum must lie in the same affine subspace as the source spectrum; a statement about the source alone would not justify the planar reduction. Toeplitz injectivity supplies precisely this additional fact. No regularity of the symbols is assumed. Arbitrary infinite-dimensional compression subspaces are not covered.

## 12. Finite POVMs with a persistent quadratic defect

Put $Q(x) = \sum_{j=1}^d x_j^2$. The next equivalence gives a route through finite semidefinite feasibility problems that does not prescribe a Walsh dilation, geometric decay, or an additive height correction.

**Proposition 12.1.** *For fixed compact convex $K \subset \mathbb{R}^d$ with compact $X = \operatorname{ex} K$, $A(K)|_X$ is not hyperrigid if and only if there are $\delta > 0$ and, for arbitrarily small $\epsilon > 0$, finite source and target lists $x_1, \dots, x_N, y_1, \dots, y_M \in X$, positive matrices $E_1, \dots, E_M \in M_N$, and a unit vector $h \in \mathbb{C}^N$ satisfying*

$$\sum_k E_k = I_N, \qquad \left\| \sum_k (y_k)_j E_k - \operatorname{diag}((x_i)_j) \right\| \le \epsilon \quad (1 \le j \le d),$$

*and*

$$\left\langle \left[ \sum_k Q(y_k) E_k - \operatorname{diag}(Q(x_i)) \right] h, h \right\rangle \ge \delta.$$

*Repeated source points are allowed. Both finite list lengths may vary, but $K, X$, and $\delta$ are fixed.*

*Proof.* First take a sequence of finite data with $\epsilon_n \to 0$. Define

$$\pi_n(f) = \operatorname{diag}(f(x_i)), \qquad \Phi_n(f) = \sum_k f(y_k) E_k.$$

These are a representation and a UCP map. Their classes define maps into the quotient C*-algebra

$$\mathcal{A} = \prod_n M_{N_n} / \bigoplus_n M_{N_n}.$$

The first remains a representation; the second is UCP. They agree on every affine coordinate, while the quadratic difference has quotient norm at least $\delta$. A faithful representation of $\mathcal{A}$ on a Hilbert space gives a nontrivial agreeing pair and hence failure of hyperrigidity. If a separable space is desired, the orbit of a witness under the maps on a countable dense family and their adjoints gives a separable reducing subspace that retains the discrepancy.

Conversely let $\pi, \Phi$ be a nontrivial agreeing pair. The coordinate defect identity implies

$$\Phi(Q) - \pi(Q) = \sum_j [\Phi(x_j^2) - \pi(x_j)^2] \ge 0,$$

and this operator is nonzero, since simultaneous vanishing makes the Stinespring compression multiplicative on the coordinate generators. Choose a unit vector $h$ with positive expectation $\delta_0$.

Let $P$ be the spectral measure of $\pi$. Partition $X$ into finitely many Borel cells $B_i$ of diameter at most $\eta$, choose $x_i \in B_i$, and form the finite-dimensional span of the nonzero vectors $P(B_i)h$. It contains $h$. The step multipliers $\sum_i f(x_i) P(B_i)$ preserve this span and give an exact finite-dimensional representation. Compress $\Phi$ to the same space. Its coordinate differences from the step representation are at most $\eta$, while its quadratic discrepancy at $h$ is at least $\delta_0 - \omega_Q(\eta)$, where $\omega_Q$ is the uniform modulus of continuity of $Q$ on $X$.

Partition the target $X$ into small Borel cells $C_k$, choose $y_k \in C_k$, and replace each effect of the compressed POVM on $C_k$ by an atom at $y_k$. The effects are positive and sum to the identity. Positivity bounds the coordinate changes by the mesh and the quadratic change by its modulus of continuity. Let both meshes tend to zero and fix $0 < \delta < \delta_0$. □

For fixed point lists and $h$, the displayed constraints are linear matrix inequalities, and the quadratic expectation is a linear objective in the effects. A single positive numerical result at nonzero $\epsilon$ is not sufficient. One must prove a uniform positive defect as $\epsilon \downarrow 0$ on a fixed compact extreme boundary. The matrix sizes necessarily become unbounded: bounded source sizes would, by compactness of the source points, unit vectors, and finite-matrix UCP maps, yield an exact finite-dimensional witness, contrary to the unique extension property of atomic representations.

## 13. Connections with other construction methods

The relevant literature spans convex geometry and normal-cone stratification, operator synthesis, potential theory, matrix convexity and semidefinite interpolation, Hardy-space compressions, and quantum-channel fixed points. The following connections have specific consequences for this problem. Results quoted here are distinguished from the arguments proved in this note. The comparison does not constitute a complete novelty audit.

13.1. **Convex geometry and operator synthesis.** Menne–Santilli's stratification theorem relates the number of independent normal directions to rectifiable strata [MS, Theorem 4.12]. This suggests separating smooth points, edge points, and vertices. One must retain exceptional sets: an arbitrary spectral measure can be concentrated on a set of zero surface measure. The countability argument in Lemma 5.4 supplies the stronger, exact statement needed here.

Shulman–Turowska connect operator support to linear operator equations and give a synthesis theorem for coefficient images of dimension at most two [ST, Theorem 11.1]. This parallels the two-dimensional normal lift in the energy argument above. It suggested searching for a way to preserve the normal label across a dilation. The final proof takes a simpler route: each remaining source point has at most one nonsmooth contact target, and the replacement in Proposition 5.8 preserves the entire UCP map. No general operator-synthesis theorem or bounded decomposition over a finite-incidence relation is assumed.

13.2. **Potential theory and noncommutative capacity.** Voiculescu relates the quasicentral modulus to nonlinear condenser capacity, with a precise correspondence for shifts on Cayley graphs [Voi]. This suggests treating a dilation defect as a quantity that must survive at arbitrarily small scales. The analogy is useful, but positive scalar capacity alone cannot provide a dilation.

There is an elementary warning already in dimension four. Surface measure on $S^3$ has finite inverse-square energy: near a point, the integral is bounded by a constant times $\int_0^1 r^{-2} r^2\, dr$. Nevertheless its coordinate system is hyperrigid. If an agreeing UCP map is defined on $C(S^3)$, then

$$\sum_{j=1}^{4} (\Phi(x_j^2) - \pi(x_j)^2) = \Phi(1) - I = 0.$$

Each summand is positive, so every coordinate is in the multiplicative domain. Thus a capacity-based construction must solve joint positivity and moment identities, not only find a measure with finite energy.

13.3. **Choquet order, matrix convexity, and finite interpolation.** Davidson–Kennedy identify the unique extension property of a multiplication representation with maximality in the dilation order [DK, Theorem 7.6]. Their dilation order differs from classical Choquet order; the distinction is exactly why ordinary barycentric transport is insufficient here. At an extreme point a classical barycentric probability kernel is necessarily a point mass.

The matrix-range framework of Davidson–Dor-On–Shalit–Solel [DDSS] and the minimal and maximal matrix convex sets of Passer–Shalit–Solel [PSS] suggest a different construction route: positive matrix-valued interpolation on increasingly fine point sets. Proposition 12.1 makes the required limiting statement precise. General scaled normal-dilation theorems do not themselves preserve an extreme represented spectrum. Similarly, a matrix convex gap at a fixed matrix size does not by itself give our diffuse counterexample.

13.4. **Quantum information and fixed observables.** Quantum-channel theory distinguishes observables fixed by a channel from those commuting with its operation elements. Magajna proves that a countable family of commuting normal Kraus operators on a separable Hilbert space forces the latter conclusion for a normal channel [Mag, Theorem 4.6]; Carbone–Jenčová

study fixed-point algebras under a faithful normal invariant state [CJ, Proposition 4]. Two elementary consequences help filter proposed constructions.

First, if a UCP channel $T$ fixes commuting self-adjoint operators $A_j$ and has a faithful invariant state $\omega$, then

$$T(A_j^2) - A_j^2 \geq 0, \qquad \omega(T(A_j^2) - A_j^2) = 0.$$

Faithfulness forces equality. The multiplicative-domain argument then shows that $T$ fixes $C^*(A_1, \dots, A_d)$. This proof does not require finite-dimensional Hilbert space. Second, within that normal-channel setting, commuting normal Kraus operators are excluded by Magajna's theorem. A channel construction therefore needs operation elements outside this commuting class and cannot have a faithful invariant state. These conditions are necessary, not a construction.

13.5. **Hardy spaces and finite multiplicity.** The Brown–Halmos displacement identity and the matrix-symbol Toeplitz/Hankel framework of Gu–Zheng [BH, GZ] make positivity of a compressed representation automatic. Exact commutativity of the compressed coordinates is a separate constraint. The flat-family argument in this note shows that simply adding two or three internal components to scalar Hardy space cannot work. Multiplicity at least four is an unexcluded case of this mechanism, not evidence of an example. Smooth or finite-band symbols may bring further restrictions that the present argument does not decide.

13.6. **Dimension and range restrictions.** Tian reduces finite-dimensional extension problems to three-dimensional operator systems by finite matrix amplification [Tia]. The generated algebra becomes $M_{3r}(A)$; it is noncommutative even when $A$ is commutative. Thus its three is a complex operator-system dimension and is not the affine dimension sought here. The tight extension results of Clouâtre–Thompson [CT] restrict the range to the represented von Neumann algebra and do not exclude arbitrary $B(H)$-valued extensions. Scherer's spectrahedral theorem [Sch2, Theorem 3.7] applies to compact spectrahedra defined by a finite matrix pencil with closed extreme boundary; no extension to their projections is used.

## 14. A nonspectrahedral four-dimensional positive example

A mixed cubic graph initially seemed to offer a simple test beyond the Walsh construction. Its geometry is explicit, and it escapes the spectrahedral theorem. A further positivity calculation nevertheless rules it out. This example is useful precisely because it separates those two obstructions.

**Proposition 14.1.** *For $t = (t_1, t_2, t_3) \in [-1, 1]^3$, put*

$$g(t) = |t|^2 + \tfrac{1}{4} t_1 t_2 t_3, \qquad K = \{(t, z) : t \in [-1, 1]^3,\ g(t) \leq z \leq 4\}.$$

*Then $K$ is a compact convex body of affine dimension four, and*

$$\operatorname{ex} K = \{(t, g(t)) : t \in [-1, 1]^3\} \cup \{(v, 4) : v \in \{-1, 1\}^3\}$$

*is closed. The body $K$ is not a spectrahedron.*

*Proof.* The Hessian of $g$ has diagonal entries 2 and off-diagonal entries $t_k/4$. The sum of absolute values of the off-diagonal entries in each row is at most $1/2$, so $\nabla^2 g \geq (3/2)I$ on the cube. In particular $g$ is strongly convex. Also $g \leq 13/4 < 4$ on the cube, so $K$ is compact and has nonempty interior.

Strong convexity makes every point of the lower graph extreme. Every point strictly between the lower graph and the top is in a nontrivial vertical segment. A top point is extreme precisely when its horizontal coordinate is a cube vertex. These observations prove the displayed formula and closedness.

Suppose $K = \{(t, z) : L(t, z) \geq 0\}$ for a finite Hermitian affine matrix pencil. At the interior point $e = (0, 0, 0, 1)$, any vector in $\ker L(e)$ lies in the common kernel of the pencil: use positivity at $e \pm \epsilon v$ for each direction $v$. Remove this common kernel, so that $L(e) > 0$. The polynomial $p = \det L$ is nonzero and vanishes on the open lower-graph patch, because positive definiteness there would make a boundary point interior. Polynomial division in $z$ now shows that

$$r(t, z) = z - |t|^2 - \tfrac{1}{4}t_1t_2t_3 \quad \text{divides } p(t, z).$$

Indeed, the remainder is a polynomial in $t$ vanishing on an open cube. Along the line $e + s(1, 1, 1, -16)$ the factor $r$ is

$$1 - 16s - 3s^2 - \tfrac{1}{4}s^3.$$

Its derivative is $-\frac{3}{4}(s + 4)^2 - 4 < 0$, so the cubic has exactly one real root, which is simple, and two nonreal roots. In contrast,

$$\det L(e + sv) = \det L(e)\, \det\bigl(I + sL(e)^{-1/2}L(v)_{\mathrm{lin}}L(e)^{-1/2}\bigr)$$

has only real zeros, because the middle matrix coefficient is Hermitian. Divisibility is a contradiction. □

This test also avoids the thin-source obstruction. Restrict the lower graph to an interior cube, and push forward three-dimensional Lebesgue measure under its normal lift. The lift is Lipschitz there and its inverse, metric projection followed by horizontal projection, is Lipschitz. The pushed-forward measure has finite inverse-square energy, by comparison with $\int_0^1 r^{-2}r^2\, dr$. This only shows that the earlier zero-capacity test does not apply.

### 14.1. **Why the cubic perturbation is still rigid.**

**Lemma 14.2.** *Let $N_1, \dots, N_d$ be self-adjoint contractions on $H \oplus L$, with blocks*

$$N_i = \begin{pmatrix} A_i & B_i \\ B_i^* & C_i \end{pmatrix}, \qquad D = \sum_{i=1}^{d} B_iB_i^*.$$

*For any word $i_1 \cdots i_k$, writing $P$ for the projection onto $H$,*

$$\|PN_{i_1} \cdots N_{i_k}|_H - A_{i_1} \cdots A_{i_k}\| \leq \binom{k}{2}\|D\|.$$

*The right-hand side is understood as zero for $k = 0, 1$.*

*Proof.* Put $d_0 = \|D\|^{1/2}$. Each $\|B_i\| \leq d_0$, and $\|[P, N_i]\| = \|B_i\|$. The commutator product rule therefore gives

$$\|PN_{i_1} \cdots N_{i_r}(I - P)\| \leq rd_0.$$

In multiplying a word of length $k - 1$ by $N_{i_k}$, its upper-left block becomes its previous upper-left block times $A_{i_k}$, plus its previous upper-right block times $B_{i_k}^*$. All $A_i$ are contractions. Thus the error bound increases by at most $(k - 1)d_0^2$. Induction proves the assertion. □

For a real polynomial $p(t) = \sum_\alpha c_\alpha t^\alpha$ define

$$C(p) = \sum_{|\alpha| \geq 2} |c_\alpha| \binom{|\alpha|}{2}.$$

Constant and linear terms contribute nothing to this quantity.

**Theorem 14.3.** *Let $X$ be a compact subset of $[-1,1]^d \times \mathbb{R}$, with coordinate functions $(t_1, \dots, t_d, z)$. Suppose*

$$z \geq g(t) := \sum_{i=1}^{d} t_i^2 + p(t) \quad \text{on } X, \qquad C(p) < 1.$$

*Every unital representation $\pi : C(X) \to B(H)$ supported on $G = \{(t,z) \in X : z = g(t)\}$ has the unique extension property relative to the affine coordinate system. No convexity hypothesis on $g$ is needed for this assertion.*

*Proof.* Let $\Phi : C(X) \to B(H)$ be unital completely positive and agree with $\pi$ on the coordinate functions. Take a Stinespring dilation $\Phi(f) = V^* \rho(f) V$, and identify $VH$ with $H$. The commuting self-adjoint contractions $N_i = \rho(t_i)$ have the blocks in Lemma 14.2, with $A_i = \pi(t_i)$. In particular the $A_i$ commute. Put

$$E = \Phi(p(t)) - p(A_1, \dots, A_d).$$

Applying that lemma to a fixed ordering of every monomial gives $\|E\| \leq C(p)\|D\|$. Since $\pi$ is supported on $G$, $\Phi(z) = \pi(z) = g(A)$. Positivity of $z - g(t)$ consequently yields

$$0 \leq \Phi(z - g(t)) = -D - E, \qquad 0 \leq D \leq -E.$$

Here $E$ is self-adjoint. Hence $\|D\| \leq \|E\| \leq C(p)\|D\|$, forcing $D = 0$ and every $B_i = 0$. The horizontal coordinates therefore intertwine: $\rho(t_i)V = V\pi(t_i)$, and the same holds for their polynomial $g$.

Set $h = z - g(t) \geq 0$. Now $\Phi(h) = 0$, so $\rho(h)^{1/2} V = 0$ and hence $\rho(h)V = 0$. It follows that

$$\rho(z)V = \rho(g(t))V = Vg(A) = V\pi(z).$$

The real coordinate functions generate $C(X)$. Their intertwining therefore proves $\Phi = \pi$ on all of $C(X)$. □

**Corollary 14.4.** *For $|\epsilon| < 1/3$, let*

$$g_\epsilon(t) = t_1^2 + t_2^2 + t_3^2 + \epsilon t_1 t_2 t_3, \qquad K_\epsilon = \{(t,z) : t \in [-1,1]^3,\ g_\epsilon(t) \leq z \leq 4\}.$$

*Then $K_\epsilon$ is a four-dimensional compact convex body with closed extreme boundary, and its affine coordinate system is hyperrigid on $C(\operatorname{ex} K_\epsilon)$. In particular this holds for the previously proposed value $\epsilon = 1/4$.*

*Proof.* The Hessian of $g_\epsilon$ has diagonal entries 2 and off-diagonal row sums bounded by $2|\epsilon|$. Thus $\nabla^2 g_\epsilon \geq (2 - 2|\epsilon|)I > 0$ on the cube. Also $g_\epsilon \leq 3 + |\epsilon| < 4$. The body is compact, convex, and has nonempty interior, and

$$\operatorname{ex} K_\epsilon = \{(t, g_\epsilon(t)) : t \in [-1,1]^3\} \ \cup\ \{(\sigma, 4) : \sigma \in \{-1,1\}^3\}.$$

Indeed, the lower graph is exposed by the strict supporting tangent planes of $g_\epsilon$; a point strictly between the lower graph and the roof lies on a nontrivial vertical segment; and the only extreme points of the roof are its eight vertices. This extreme boundary is compact.

For $p(t) = \epsilon t_1 t_2 t_3$, Theorem 14.3 applies because $C(p) = 3|\epsilon| < 1$. Thus every representation supported on the lower graph has the unique extension property. Each of the eight remaining evaluations has that property because its point is extreme. Their arbitrary multiplicities also have the property, and the unique extension property is preserved by direct sums. Decomposing an arbitrary representation along the lower graph and the eight roof points proves the corollary. □

For the cubic alone, the controlling identity is particularly short:

$$\Phi(t_1 t_2 t_3) - A_1 A_2 A_3 = A_1 B_2 B_3^* + B_1 B_2^* A_3 + B_1 C_2 B_3^*.$$

Each summand has norm at most $\|D\|$. This gives the constant 3 directly. The result makes the cubic body a positive example beyond the finite-spectrahedron class, by Proposition 14.1; it is not a counterexample candidate.

## 15. The remaining low-dimensional questions

The lower-dimensional obstruction is now complete:

$$4 \leq d_{\min} \leq 33.$$

No construction in affine dimension four is obtained here. The signing and proximal arguments exclude direct modifications of the stationary geometric Walsh model; the correlated expectation obstruction and the new rank-one, Hardy, and polynomial results exclude other simple mechanisms. These are restrictions on particular constructions, not evidence that all four-dimensional bodies are hyperrigid.

Proposition 12.1 suggests a different route: fix a compact extreme boundary in dimension four and seek finite POVMs whose affine operator-norm errors vanish while a positive quadratic discrepancy persists. Both the boundary and the positive discrepancy must stay fixed as matrix sizes grow. The cubic body in Proposition 14.1 is excluded by Corollary 14.4, despite its nonspectrahedral geometry. A different body must escape that polynomial estimate.

A channel construction must likewise avoid a faithful invariant state and the commuting normal Kraus class discussed above. Hardy compression has first unexcluded multiplicity four. In either approach, exact commutation, extremality of the represented spectrum, and a nonzero coordinate defect must all be verified; none follows from the other two.

**Preparation and verification.** This manuscript was developed and edited with substantial language-model assistance. Separate automated review tasks rederived the compression identities, checked the geometric and measure-theoretic arguments, and tested proposed constructions against the stated hypotheses. In version two, a fresh adversarial review rederived the dilation replacement and replaced the residual disintegration with bounded Borel endpoint interpolation. The source decomposition, selectors, and nonseparable reduction were checked separately. Exact rational arithmetic verifies the numerical certificate in Section 6. These checks are not independent human mathematical verification and are not a formal proof certification. The bibliography identifies the source versions used; the limited literature comparison does not establish priority for every subsidiary lemma. The presentation separates complete arguments from the open questions above.

## Appendix A. Normal-lift capacity is not a necessary condition

The normal lift gives a convenient metric for the supporting-gap criterion, but it need not give the strongest one. The next example separates the two tests. Its represented source has the unique extension property even though its normal lift supports a measure of finite inverse-square energy.

**Theorem A.1.** *There are a compact convex body $K \subset \mathbb{R}^4$ with closed extreme boundary and a compact set $Z \subset \operatorname{ex} K$ such that:*

(i) *every point of $Z$ is exposed and has a unique unit outward normal;*
(ii) *the normal lift of $Z$ supports a probability measure of finite inverse-square energy;*
(iii) *every representation concentrated on $Z$ has UEP relative to $A(K)|_{\operatorname{ex} K}$.*

*Proof.* We construct a thin Cantor set whose normal field records two additional independent collections of digits. For signs $\omega_n \in \{-1,1\}$, $n \ge 0$, put

$$s(\omega) = \left( \sum_{n\ge0} 4^{-n}\omega_n, 10^{-3}\sum_{n\ge0} 64^{-n}\omega_n, 10^{-3}\sum_{n\ge0} 128^{-n}\omega_n \right), \qquad C = s(\{-1,1\}^{\mathbb{N}_0}).$$

The first coordinate is an injective Cantor expansion, so $s$ is a homeomorphism onto $C$. Its first three vector coefficients are linearly independent by the Vandermonde determinant; hence $\operatorname{aff} C = \mathbb{R}^3$. Also $C \subset (-2,2)^3$ and $|s| < 2$.

If two sequences first differ at index $n$, their first-coordinate difference is at least $(4/3)4^{-n}$, whereas their second- and third-coordinate differences are at most $2\cdot 10^{-3}64^{-n}/(1-1/64)$ and $2\cdot 10^{-3}128^{-n}/(1-1/128)$, respectively. In particular,

$$|s_j - t_j| \le \frac{8}{7000}|s_1 - t_1|^2 \quad (j=2,3), \qquad |s-t|^2 \le 2|s_1-t_1|^2. \tag{A.1}$$

Define continuous functions $a, b : C \to [1,2]$ by

$$a(s(\omega)) = 1 + \sum_{k\ge0} 2^{-(k+1)}\frac{1+\omega_{3k+1}}{2}, \qquad b(s(\omega)) = 1 + \sum_{k\ge0} 2^{-(k+1)}\frac{1+\omega_{3k+2}}{2}.$$

Set $p(s) = 2s + (0, a(s), b(s))$. The preceding estimates imply

$$\big|\langle p(s) - 2s, t-s\rangle\big| \le \frac{32}{7000}|t-s|^2 < \frac{1}{2}|t-s|^2 \qquad (s,t \in C). \tag{A.2}$$

Thus the data $g(s) = |s|^2$, $\nabla g(s) = p(s)$ admit the strongly convex extension

$$g(x) = \max_{s\in C}\left\{ |s|^2 + \langle p(s), x-s\rangle + \tfrac{1}{2}|x-s|^2 \right\}.$$

Indeed, subtracting $|x|^2/2$ leaves a supremum of affine functions. For $x \in C$, (A.2) shows that the unique maximizer is $s = x$, and $g(x) = |x|^2$.

For clarity, this uniqueness also proves differentiability at every $x \in C$. If $s_h$ maximizes at $x+h$, compactness and continuity force $s_h \to x$. Let $Q_s$ denote the quadratic inside the maximum. Then

$$Q_x(x+h) - Q_x(x) \le g(x+h) - g(x) \le Q_{s_h}(x+h) - Q_{s_h}(x).$$

Both bounds are $\langle p(x), h\rangle + o(|h|)$.

Choose $M > \max_{[-2,2]^3} g$ and let

$$K = \{(x,z) : x \in [-2,2]^3,\ g(x) \le z \le M\}, \qquad Z = \{P_s = (s, |s|^2) : s \in C\}.$$

Strict convexity shows that the full extreme boundary consists of the lower graph and the eight vertices of the top face, exactly as in Lemma 6.5. It is compact, and $K$ has nonempty interior. At $P_s$, the supporting plane is strict away from $P_s$, and the unique outward unit normal is

$$n(s) = \frac{(p(s), -1)}{\sqrt{1+|p(s)|^2}}.$$

We next show that the lift $P_s + n(s)$ carries finite energy. The paired lift $(P_s, n(s))$ and the sum lift $P_s + n(s)$ are bi-Lipschitz equivalent: the supporting-normal inequalities give

$$|P_s - P_t|^2 + |n(s) - n(t)|^2 \le |P_s + n(s) - P_t - n(t)|^2 \le 2\big(|P_s - P_t|^2 + |n(s) - n(t)|^2\big).$$

By (A.1), the paired lift is in turn bi-Lipschitz equivalent to the coordinates $(s_1, a(s), b(s))$. In one direction all its components are Lipschitz functions of these three coordinates. In the other, recover $s$ from the position and then recover $p(s)$ from the normal by division by its last coordinate, which is bounded away from zero.

Under product measure on the signs, $a$ and $b$ are independent uniform variables on $[1,2]$. Conditional on the digits defining them, the remaining contribution to $s_1$ is a translate of

$$\sum_{k\ge 0} 64^{-k}\omega_{3k}.$$

Its probability in an interval of length $2r$ is bounded by a fixed multiple of $r^{1/6}$. To verify this, choose $N$ with $64^{-N} \le r < 64^{-(N-1)}$. An interval of that length meets at most two level-$N$ cylinders, each of mass $2^{-N}$; the separation follows by comparing the first differing digit with the geometric tail. Conditioning and integrating over $a, b$ therefore give a ball bound of the form $Cr^{2+1/6}$ for $(s_1, a, b)$, uniformly in the center. The same bound holds on the normal lift by bi-Lipschitz equivalence. For its pushed-forward probability measure $\nu$, it follows that

$$\iint |u-v|^{-2}\, d\nu(u) d\nu(v) \le 1 + 2C \int_0^1 r^{-3} r^{13/6}\, dr < \infty.$$

Finally, consider the actual supporting affine functions

$$b_{P_s}(x,z) = z - |s|^2 - \langle p(s), x - s\rangle.$$

They expose $P_s$, since $b_{P_s}(x,z) \ge z - g(x) + |x-s|^2/2$ on $K$. On the represented source, (A.1)–(A.2) give

$$0 \le b_{P_s}(P_t) \le 3|s_1 - t_1|^2.$$

The first coordinate lies in the compact interval $[-4/3, 4/3]$, which supports no nonzero measure of finite inverse-square energy. Theorem 3.1 proves the unique extension property on $Z$. □

## Appendix B. Varying distributions and uniformly positive defects

The local defect estimate in Lemma 2.3 imposes a restriction independent of the choice of convex graph. Here is its consequence for varying three-point distributions.

At coordinate $n$, let the symmetric distribution on $\{0, \pm R_n\}$ have second moment $b_n$, where $1 < b_n < R_n^2$. Put $a_n = b_n^{-1}$. Use the branch spaces of the Walsh construction, with these varying tail distributions. Replace the formula for $V$ by

$$V\chi_F = \left(\prod_{n\in F} \sqrt{a_n}\right) \chi_F^{(0)} + \sum_{k\in F} \sqrt{1-a_k} \left(\prod_{n\in F_{>k}} \sqrt{a_n}\right) \chi_{F\setminus\{k\}}^{(k)},$$

where the normalized branch monomial is $\chi_F^{(k)} = (\prod_{n\in F_{>k}} b_n^{-1/2}) \prod_{n\in F} x_n$. The proof of Lemma 6.2 remains valid because

$$\prod_{n\in F} a_n + \sum_{k\in F} (1-a_k) \prod_{n\in F_{>k}} a_n = 1.$$

In particular all square-free products compress exactly. For the coordinate-square defect, parity and the same branch calculation give

$$(\Phi(x_n^2) - I)\chi_F = \left(\prod_{i\in F_{>n}} a_i\right) \begin{cases} b_n - 1, & n \notin F, \\ R_n^2/b_n - 1, & n \in F \end{cases} \chi_F. \tag{B.1}$$

**Theorem B.1.** *Assume $\sum_n (b_n - 1) < \infty$. Let $w_n \in \mathbb{R}^D$ satisfy $\sum_n R_n |w_n| < \infty$, with at least one $w_n \ne 0$, and put*

$$t = \sum_n w_n x_n, \qquad s = \sum_n w_n \varepsilon_n.$$

*There is no compact convex body with compact extreme boundary carrying an affine-agreeing pair $(\pi, \Psi)$ for which the first $D$ ambient coordinate functions $\tau_j$ satisfy $\pi(\tau_j) = M_{s_j}$ and*

$$\Psi(\tau_j^2) \geq \Phi(t_j^2) \qquad (1 \leq j \leq D).$$

*In particular, no positive barycentric-kernel construction realizes these data, regardless of its additional coordinates.*

*Proof.* The product $a_* := \prod_n a_n$ is positive. Equation (B.1) gives

$$\Phi(x_n^2) - I \geq a_* \min\{b_n - 1, R_n^2/b_n - 1\} I > 0.$$

Uniform convergence and exact compression of mixed products imply

$$\Phi(t_j^2) - M_{s_j}^2 = \sum_n (w_n)_j^2 (\Phi(x_n^2) - I).$$

For some $j$, a nonzero summand bounds this defect below by a positive multiple of $I$. The asserted inequality for $\Psi$ contradicts Lemma 2.3. For a barycentric kernel it follows from scalar Jensen's inequality for a coordinate square and positivity of compression. □

The same obstruction applies to finite nonlinear coordinates.

**Proposition B.2.** *Under the preceding summability hypothesis, every nonconstant real multiaffine polynomial $p$ in finitely many coordinates satisfies*

$$\Phi(p^2) - M_{p(\varepsilon)}^2 \geq c_p I$$

*for some $c_p > 0$. It therefore cannot be used as an affine coordinate of a compact-extreme-boundary barycentric construction.*

*Proof.* Exact square-free compression identifies the defect with $T^*T$, where $T = M_p V - V M_{p(\varepsilon)}$. Restrict this error to the initial branch $Y_0$. If $p$ involves the first $N$ variables, its tail factor is bounded below by the positive product $\prod_{n>N} b_n^{-1/2}$. It suffices to prove injectivity of the finite-prefix error.

Identify functions on the binary cube with multiaffine polynomials. Let $\mathcal{D}$ scale a monomial $x_F$ by $\prod_{i \in F} b_i^{-1}$. If the error vanishes on a vector represented by $h$, then, with $q = \mathcal{D}h$,

$$p(x)q(x) = \mathcal{D}(p(\varepsilon)h(\varepsilon))(x)$$

on the full three-point product grid. Both sides have degree at most two in each variable, so this is a polynomial identity. The right side is multiaffine. Whenever $p = p_0 + x_i p_1$ with $p_1 \neq 0$, write $q = q_0 + x_i q_1$; comparison of the $x_i^2$ coefficient gives $p_1 q_1 = 0$, hence $q_1 = 0$. Thus $q$ and $h$ are independent of every active variable of $p$. The identity reduces to

$$[p(x) - p(x_1/b_1, \ldots, x_N/b_N)]q(x) = 0.$$

The first factor is nonzero, because $p$ is nonconstant and every $b_i > 1$. Hence $q = 0$. The finite-prefix error has positive minimum singular value; tensoring with the bounded-below tail proves the claim. □

## Appendix C. Correlated magnitudes and an expectation obstruction

Independence of the magnitudes is not essential to the Walsh isometry. This observation gives a distinct family of exact compression maps. A simple expectation test, however, excludes an entire range of its possible geometric realizations.

Fix $b > 1$ and $0 < \delta < b$. Let $\theta$ be uniform on the circle and let $\xi_n$ be independent uniform signs, independent of $\theta$. Put

$$x_n = \xi_n \sqrt{b + \delta \cos(4^n \theta)}. \tag{C.1}$$

These coordinates share their angle and are not independent. Nevertheless, if $S_n = x_n^2$, then for every finite index set and integers $0 \le k_n \le 3$,

$$\mathbb{E}\prod_n S_n^{k_n} = \prod_n \mathbb{E}S_n^{k_n}. \tag{C.2}$$

Indeed, a nonconstant term in the Fourier expansion has frequency $\sum_n j_n 4^n$ with $|j_n| \le 3$. Its largest nonzero term cannot be cancelled, since $4^m > 3\sum_{n<m} 4^n$. Thus only products of constant Fourier coefficients survive. By contrast,

$$\mathbb{E}[(S_n - b)^4(S_{n+1} - b)] = \delta^5/16 \neq 0,$$

as follows from the $\cos 4t$ coefficient in $\cos^4 t$.

On the initial branch use (C.1) at every coordinate; on branch $Y_k$ use the binary prefix, the zero-valued star, and the same correlated law on its tail. The angle and all prefix and tail signs are independent. For compactness, use disjoint tagged copies of the binary symbols and of the compact tail intervals, even where their numerical values agree. Each fixed branch is a compact image of its angle and sign space; branches with stars tending to infinity accumulate only on the binary source. Thus the union is compact, $Y_0$ is clopen, and every numerical coordinate is continuous.

**Proposition C.1.** *With these branch laws and the same formula for $V$ as in (6.1), using $a = b^{-1}$, the map $V$ is an isometry and every square-free product compresses exactly. Moreover $V1 = \mathbf{1}_{Y_0}$ and*

$$(\Phi(x_n^2) - I)\chi_F = b^{-|F_{>n}|}\begin{cases} b-1, & n \notin F, \\ \delta^2/(2b^2), & n \in F \end{cases}\chi_F. \tag{C.3}$$

*In particular each coordinate-square defect is positive and injective.*

*Proof.* Independent signs make different monomial labels orthogonal. Equation (C.2) gives the same normalized monomial norms as the independent law. A surviving square-free matrix coefficient uses only exponents zero and two, so the complete proof of Lemma 6.2 applies. For the square defect, $\mathbb{E}x_n^4 = b^2 + \delta^2/2$. If $n \in F$, the total pre-star branch weight is $b^{-|F_{>n}|-1}$ and the weighted second moment is $(b^2 + \delta^2/2)/b$; the star contributes zero. Subtracting $I$ gives the second coefficient in (C.3). If $n \notin F$, the calculation uses the ordinary second moment $b$, giving the first. □

**Theorem C.2** (An obstruction independent of the height correction)**.** *For the correlated model, let $w_n \in \mathbb{R}^D$ be a nonzero absolutely summable sequence and put*

$$t = \sum_n w_n x_n, \qquad s = \sum_n w_n \varepsilon_n, \qquad C = s(G).$$

*Suppose a continuous height $z$ satisfies*

$$\Phi(z) = M_{|s|^2}, \qquad z(y) \ge |t(y)|^2 - \operatorname{dist}(t(y), C)^2 \quad (y \in Y_0).$$

*Then*

$$\frac{1}{2\pi}\int_0^{2\pi} \sqrt{b + \delta\cos\theta}\, d\theta \le 1. \tag{C.4}$$

*No additive form for $z$, and no geometric decay of $|w_n|$, is assumed.*

*Proof.* Let $L = \sum_n |w_n|^2 > 0$ and denote the mean in (C.4) by $m$. Exact height compression and $V1 = \mathbf{1}_{Y_0}$ give $\mathbb{E}_{Y_0} z = L$, whereas independent signs give $\mathbb{E}_{Y_0}|t|^2 = bL$. For each branch point,

compare $t$ with the source point having the same signs. Averaging first over signs and then over the angle gives

$$\mathbb{E}_{Y_0}\operatorname{dist}(t,C)^2 \le \mathbb{E}\left|\sum_n w_n\xi_n\big(\sqrt{b+\delta\cos(4^n\theta)}-1\big)\right|^2 = (b+1-2m)L.$$

Containment gives the reverse lower bound $(b-1)L$. Dividing by $L$ proves $m \le 1$. Absolute uniform convergence justifies all expansions. □

**Corollary C.3.** *The distribution $b = 6/5$, $\delta = 1$ admits no such exact height in any finite horizontal dimension. Suppose a body $K$ satisfies the geometric hypotheses of Theorem 8.3, contains every encoded point $(t(y), z(y))$, and a positive barycentric kernel realizes the affine-agreeing map on the signed-source representation. Then no finite strictly convex graph can repair this model when $D \le 3$.*

*Proof.* The two chords of the square-root function on $[b-\delta, b]$ and $[b, b+\delta]$ give

$$m \ge \sqrt{b} + \frac{\sqrt{b+\delta}+\sqrt{b-\delta}-2\sqrt{b}}{\pi} > \frac{\sqrt{b}+\sqrt{b+\delta}+\sqrt{b-\delta}}{3}.$$

The last inequality uses strict concavity and $\pi > 3$. At the stated parameters, the three square roots exceed 109/100, 148/100, and 44/100, respectively. Thus $m > 301/300 > 1$. For the graph assertion, exact mixed products and (C.3) give an injective defect in any nonzero horizontal coordinate. A positive barycentric kernel preserves that lower bound, and the signed source has full spectral support. Theorem 8.3 forces the excluded quadratic envelope. □

More generally, the mean-radius condition forces $1 < b < \pi^2/8$ and $\delta^2 > 4(b-1)$. For the first inequality, the mean is strictly decreasing in $\delta$ and its limit at $\delta = b$ is $2\sqrt{2b}/\pi$. For the second, its value is strictly larger than the endpoint chord average $(\sqrt{b+\delta}+\sqrt{b-\delta})/2$. These restrictions leave admissible pairs near $(1,1)$; they do not exclude the whole correlated family or establish containment for any remaining pair.

## Appendix D. Restrictions on direct operator constructions

We record two short diagnostics that are independent of the Walsh construction. They explain why two natural direct constructions cannot supply the desired diffuse example. Neither is a universal dimension bound.

**Proposition D.1.** *Suppose an affine-agreeing POVM $E$ satisfies $E(B) \le \nu(B)I$ for every Borel $B \subset X$, where $\nu$ is a finite positive scalar measure. Then the spectral support of $\pi$ has at most $\lfloor \nu(X) \rfloor$ points and $E = P$. In particular, an operator-valued density with integrable operator norm cannot yield a diffuse counterexample.*

*Proof.* For a spectral-support point $x$, choose unit vectors $h_n \in P(B(x,1/n))H$. Their scalar measures for $E$ are probabilities dominated by $\nu$, with barycenters tending to $x$. Any weak limit is $\delta_x$ by extremality, and remains dominated by $\nu$. Therefore $\nu(\{x\}) \ge 1$. There are at most $\lfloor \nu(X) \rfloor$ such points, and Lemma 2.2 gives equality of the maps. For a positive density $F$ relative to a scalar measure $\mu$, take $d\nu = \|F\|\, d\mu$. □

This is an order-domination hypothesis, not merely absolute continuity with respect to a scalar control measure. It does not automatically apply to a correction of a spectral measure.

**Proposition D.2.** *Let $\rho : C(X) \to L^\infty(\mathbb{T})$ be a unital $*$-representation, and let $P_+$ be the orthogonal projection from $L^2(\mathbb{T})$ onto the scalar Hardy space $H^2$. The compression*

$$\Phi(f) = P_+ M_{\rho(f)}|_{H^2}$$

*cannot be a nontrivial UCP map agreeing with a representation on* $A(K)|_X$.

*Proof.* Let $S$ be the unilateral shift on $H^2$ and let $e_0$ be its first basis vector. For self-adjoint Toeplitz operators $A, B$, put $u_A = S^*Ae_0$ and $u_B = S^*Be_0$. The identities $S^*AS = A$ and $SS^* = I - e_0e_0^*$ give

$$S^*[A, B]S - [A, B] = u_Au_B^* - u_Bu_A^*.$$

If $A, B$ commute and $u_A \neq 0$, the two rank-one products agree. It follows that $u_B = \lambda u_A$ for some real $\lambda$. For a real symbol $f$, $u_{T_f}$ consists of its strictly positive Fourier coefficients. The negative coefficients are their conjugates. Consequently, commuting real scalar Toeplitz symbols are all of the form $\lambda_j f + c_j$ for one real symbol $f$, unless all are constant.

In an affine-agreeing compression the compressed coordinates commute, because they equal the coordinates of the representation. Their joint spectral support therefore lies on an affine line, or is a singleton. A line meets $\operatorname{ex} K$ in at most two points: every interior point of its intersection segment with $K$ is nonextreme. The represented support is thus finite, and Lemma 2.2 applies. □

In contrast, small norm and low Kraus rank do not themselves impose rigidity. Any counterexample $\Psi(f) = V^*\rho(f)V$ can be amplified on $H \oplus L \oplus L \oplus \cdots$ to $\widehat{\pi} = \pi \oplus \rho \oplus \rho \oplus \cdots$. The isometry

$$W(h, \ell_1, \ell_2, \ldots) = (0, Vh, \ell_1, \ell_2, \ldots)$$

satisfies $W^*\widehat{\pi}(f)W = \Psi(f) \oplus \rho(f) \oplus \rho(f) \oplus \cdots$. Mixing this compression with the identity gives a nontrivial two-Kraus channel fixing the affine coordinates and arbitrarily close to the identity. This is a reformulation of a counterexample, rather than an independent construction of one.

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